# WATER STORAGE CHANGE AND CLIMATE ATTRIBUTION ACROSS THAILAND

by

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## Abstract

Thailand requires basin-scale information on water storage, but climatic variability, water management and measurement uncertainty overlap in satellite observations. We analysed terrestrial water storage anomalies (TWSA) and derived groundwater storage anomalies (GWSA) across all 22 principal basins. CSR GRACE and GRACE-FO observations were combined with a full-compartment groundwater balance, paired naturalized and historical WaterGAP simulations, chronological climate-model evaluation, GLDAS land-storage comparisons, groundwater levels and independent reservoir observations. National seasonal Sen slopes were −7.51 mm/year for TWSA through September 2024 and −6.84 mm/year for GWSA through September 2023. Basin trend ranks agreed weakly between processing chains. A meteorological ensemble reconstructed its naturalized parent with held-out 2016–2023 NSE of 0.981 for TWSA and 0.961 for GWSA. These scores describe model reconstruction, not validation of an observed natural counterfactual. At a six-month ONI lead, season-, trend- and DMI-adjusted national correlations were −0.536 and −0.528 for naturalized TWSA and GWSA, compared with −0.142 and −0.136 for their observation-minus-baseline residuals. No basin association passed the conservative multiple-comparison screen; the short record limits statistical power. Groundwater agreement was weak and sensitive to station selection. Independent reservoir storage supported one surface-water component of the TWS residual, with adjusted r = 0.439 across 73 scene-supported months. The study distinguishes a detectable national decline from less certain basin rankings and human attribution. Its contribution is an explicit assessment of attribution credibility, supported by reproducible code and an interactive basin decision-support system.

## Contents

1 Introduction — 4

2 Background — 5

3 Problem statement — 6

4 Objectives — 7

5 Study area — 8

6 Data collection — 9–10

7 Methodology — 11–14

8 Results — 15–44

9 Conclusion — 45

10 Discussion — 46–47

11 Further work — 48–49

References — 50

Results occupy 30 of the 50 report pages, or 60%. Each results page states the comparison, the main finding and its interpretation. All 22 basins are retained in the maps, temporal panels and full-precision numerical tables.

# 1 Introduction

Water storage integrates rainfall, evapotranspiration, runoff and human water use. A rainfall deficit can reduce soil moisture, river storage and groundwater recharge at the same time that irrigation demand increases. Storage decline in such a year can therefore reflect several connected processes. For Thailand, this distinction affects how satellite observations should inform drought preparation, reservoir operation and groundwater monitoring. A map of declining storage is useful evidence of water loss, but the map alone does not identify its cause.

GRACE and GRACE-FO observe broad changes in terrestrial water mass. Their anomalies include groundwater, soil water, surface water, canopy water and snow. Groundwater must be inferred by subtracting the other compartments. The resulting GWSA inherits errors from both the gravity solution and the component estimates. Spatial averaging reduces some noise but cannot create information below the native observational support. Thailand's principal basins are useful reporting units even when their boundaries do not define independent gravimetric observations.

This study asks three connected questions. First, how do TWSA and GWSA vary across Thailand's 22 principal basins and through time? Second, how well can meteorological information reconstruct a naturalized model baseline, and what evidence supports interpreting the remaining residual? Third, how can these results be made accessible and reproducible through a decision-support website? The objectives deliberately distinguish storage detection, model evaluation and causal attribution.

## 2 Background

Previous research in the Chao Phraya has related storage dynamics to hydroclimatic extremes, demonstrating the regional value of GRACE while leaving questions about wider basin coverage and attribution (Abhishek, Kinouchi and Sayama, 2021). Chueasa, Humphries and Waqas (2024) examined ENSO and station rainfall in Northeast Thailand. That work motivates a lag-sensitive storage analysis, but a rainfall relationship cannot simply be transferred to groundwater or treated as a national causal coefficient.

The 2025 International Journal of Digital Earth article "Separating climate and human induced terrestrial water storage anomalies with GRACE data and hydrological models" provides a useful reconstruction-and-residual framework and clear multi-panel figure design. We use its visual approach for original Thailand results, while treating the residual as a mixture of human effects and discrepancy. Its use of mean absolute deviation motivates a variability diagnostic; such ratios are not complementary fractions of causal responsibility.

Sharifi et al. (2025) show why non-groundwater compartments should have a spatial response compatible with GRACE before subtraction. WaterGAP supplies a physically interpretable naturalized/historical pair (Müller Schmied et al., 2024), but agreement with its own emulator cannot establish observational accuracy. MSEA-Res supplies independently derived reservoir observations (Mahto, Fatichi and Galelli, 2025), allowing a partial surface-water check. The research gap is therefore an integrated assessment of what each source can identify, rather than another unrestricted claim that observed storage minus a model equals human water use.

## 3 Problem statement

Thailand needs information on where water storage is changing and whether the change is mainly consistent with climate variability or water management. Rainfall deficits, reduced recharge, reservoir operations and abstraction can occur together. A declining storage series therefore does not identify its cause without additional evidence.

GRACE observes total water mass over broad areas. It does not separately observe groundwater or human water use. Groundwater storage must be derived by removing the other storage compartments. Small-basin results also share gravity information, while the supporting model and monitoring records have different spatial and temporal coverage.

The central problem is that a large observation-minus-climate residual can be mistaken for measured human depletion. This study tests that interpretation rather than assuming it. A convincing assessment must distinguish model reconstruction skill, agreement with matching GLDAS components, and independent checks using groundwater levels and reservoirs.

The practical need is a clear basin assessment that shows both the storage signal and the strength of its supporting evidence. The report therefore gives equal attention to coverage, product sensitivity and failed checks, and presents these limitations alongside usable maps and time series.

## 4 Objectives

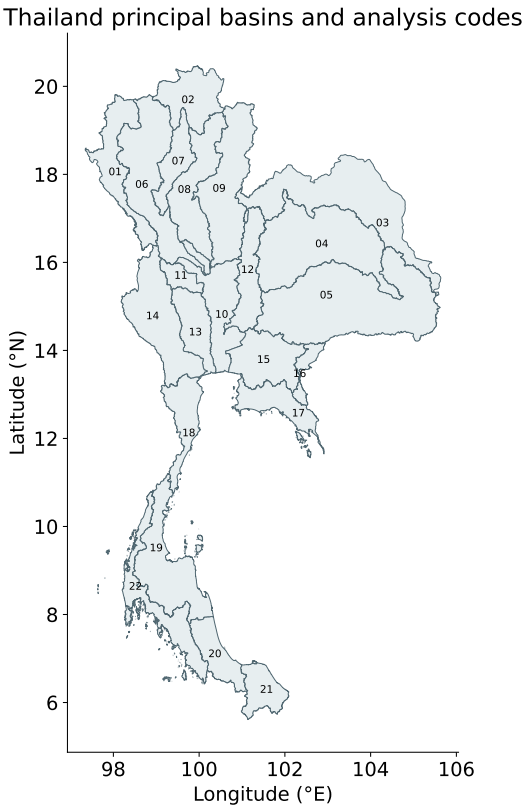
Objective 1: quantify terrestrial and groundwater storage anomalies for every principal basin in Thailand. Describe monthly variability, estimate seasonal Sen slopes, apply Mann–Kendall tests and show the results spatially and through time.

Objective 2: reconstruct a naturalized climate baseline from meteorological information and climate memory. Evaluate the reconstruction chronologically, compare matching land compartments with GLDAS, calculate the residual, and examine what groundwater levels, reservoirs and ENSO diagnostics can support about its interpretation.

Objective 3: provide a clear interactive basin decision-support website with downloadable figures, tables and code that a professor or another researcher can run.

The research questions are whether national decline is consistent across processing chains, whether weather memory improves reconstruction, and whether independent observations support the residual. The study does not assume that a high model score validates human-caused storage loss.

# 5 Study area




**Thailand principal basins**. The 22 ONWR basins are reporting polygons. Adjacent basins share broad GRACE support.

The domain comprises Thailand and all 22 principal river basins represented by the supplied Office of the National Water Resources boundary layer. The analysis covers the northern headwater basins, central plains, northeastern plateau, eastern drainage systems and peninsular basins. Basin codes are stable join keys across figures, CSV tables and the website. National values are aggregated from represented spatial support rather than calculated as an unweighted average of basin means.

All principal basins fall below CSR's approximate 200,000 km² small-area caution. The national footprint intersects 68 native mascons; inverse weight concentration is approximately 50.1, which describes spatial weighting and does not count independent observations. Adjacent basin maps share broad gravity support. Interpretation should emphasize coherent regional patterns and monitoring priorities rather than precise local aquifer attribution.

## 6 Data collection

| Data | Used for | Key limitation |
|---|---|---|
| CSR GRACE / GRACE-FO | Observed total storage | Broad gravity support |
| GLDAS Noah | Soil canopy and snow | Partial land storage only |
| G3P surface water | Groundwater subtraction | Component uncertainty |
| WaterGAP paired runs | Naturalized / historical storage | Model-defined counterfactual |
| Native P and potential ET | Climate-model predictors | Forcing dependence |
| CHIRPS rainfall | Alternative-forcing test | Different rainfall estimation |
| GLEAM MODIS ESA CCI | Hydroclimatic context | Unequal product coverage |
| GROW groundwater levels | Groundwater-state evidence | Local and mostly annual |
| MSEA-Res Level 1 | Reservoir component evidence | Sparse scene observations |
| ONI RONI v5 and DMI | ENSO / IOD diagnostics | Short common record |

All products retain their original physical definition; a flux, a storage anomaly and a groundwater level are not interchangeable.

Independent supporting inputs include CHIRPS rainfall, GLEAM and MODIS actual evapotranspiration, ESA CCI surface and root-zone moisture, MODIS land-surface temperature, GROW well observations and MSEA-Res reservoir storage. NOAA ONI, historical RONI v5 and HadISST-based DMI are frozen locally for the new teleconnection analysis. File hashes, download URLs and index versions are retained. Data coverage varies by source, and each comparison uses its own eligible paired dates.

## 6 Data collection and preparation

The input collection includes downloaded source products, provider documentation, frozen climate indices and file checksums. The original and extended GitHub packages retain the input subsets needed for a complete numerical rerun. The new ENSO extension can also be run from verified derived hydrological inputs without credentials or additional downloads.

Storage is converted to millimetres of equivalent water depth. CSR centimetres are multiplied by ten. Soil, surface, canopy and snow water are referenced consistently to the available 2004–2009 mean. Native rainfall and potential ET rates in mm/day are converted to monthly totals using the actual length of each month when reporting flux summaries.

Basin averages use represented cell area and valid data coverage. Groundwater subtraction requires the relevant components on common valid support. Cross-product comparisons use paired dates. Missing GRACE months, missing well years and absent reservoir scenes are retained as missing; they are not replaced with zero or smoothed into observations.

The primary total-storage record extends from April 2002 to September 2024. Groundwater ends in September 2023 because of component availability. The common ENSO analysis uses 2003–2023. Climate-model testing uses 2016–2023. These differing periods are stated at the relevant result rather than hidden in one common label.

## 7 Methodology storage and trend estimation

Step 1 harmonizes monthly observations, model outputs, units, reference periods and valid spatial support. Step 2 calculates groundwater storage anomalies using GWSA = TWSA − SMSA − SWSA − CWSA − SWEA. In words, groundwater equals total storage minus soil, surface, canopy and snow storage.

Step 3 estimates a seasonal Sen slope for each basin. It compares the same calendar month across years, reducing distortion by the annual cycle. A slope of −5 mm/year means the anomaly decreases by about 5 mm of equivalent water depth per year over that record. It does not measure the water remaining in an aquifer.

The seasonal Mann–Kendall test evaluates ordered change. Basin probabilities are adjusted for multiple comparisons using false discovery rate. Original probabilities, sample sizes and conditional slope intervals are retained. Serial dependence is examined through annual and resampling sensitivities; a small probability is not proof of a human cause.

National results and processing-chain comparisons provide scale context. All basins remain reporting units below the provider's approximate small-area caution. Resampling intervals are conditional on the chosen products and dates, so product disagreement is reported separately.

## 7 Methodology climate reconstruction

The climate baseline is naturalized WaterGAP storage, which excludes the direct human activities represented in its paired historical run. The statistical climate model is an emulator: it reconstructs that baseline from precipitation, potential evapotranspiration, season and weather memory. It is not a global climate projection model.

Ridge, ElasticNet, a spline model and random forest are evaluated and combined using validation performance. Past rainfall and PET help represent delayed storage responses. GRACE storage and groundwater level are excluded from the predictors, preventing those observations from directly teaching the emulator the desired residual.

The chronology is fixed: 1981–1990 warm-up, 1991–2009 training, 2010–2015 validation and 2016–2023 testing. Transformations and tuning use earlier data only. The test period does not choose model weights or ENSO-specific fits. A controlled Ridge comparison tests memory, while a frozen CHIRPS replacement tests forcing sensitivity.

RMSE expresses reconstruction error in mm. Correlation describes timing. NSE compares squared error with a target-mean benchmark: 1 is perfect, 0 equals the mean benchmark and negative is worse. Anomaly NSE removes the training monthly climatology so that seasonal agreement alone cannot dominate the interpretation.

## 7 Methodology residual and GLDAS comparison

Step 4 calculates the residual as observed storage minus the naturalized climate baseline. The word residual means the part left after subtraction; it does not identify why the difference exists.

The paired historical model provides a separate modeled human effect: historical minus naturalized storage. The remaining unexplained gap is observed minus historical storage. Therefore residual = modeled human effect + unexplained gap, or $R = H + E$. This exact identity is checked on matched months.

The unexplained gap can include forcing error, model structure, gravity error, component subtraction and unrepresented management. A large negative residual can be consistent with human depletion, but it is not an independently observed pumping record. Additive component trends use calendar-month fixed-effect OLS, because seasonal Sen slopes are not algebraically additive.

Step 5 benchmarks GLDAS only against matching soil, canopy and snow storage. Groundwater and open surface water are excluded. This tests land-component consistency, not full total storage or human effects. GLDAS also enters groundwater subtraction, so groundwater-related comparisons are partly dependent.

## 7 Methodology validation ENSO and uncertainty

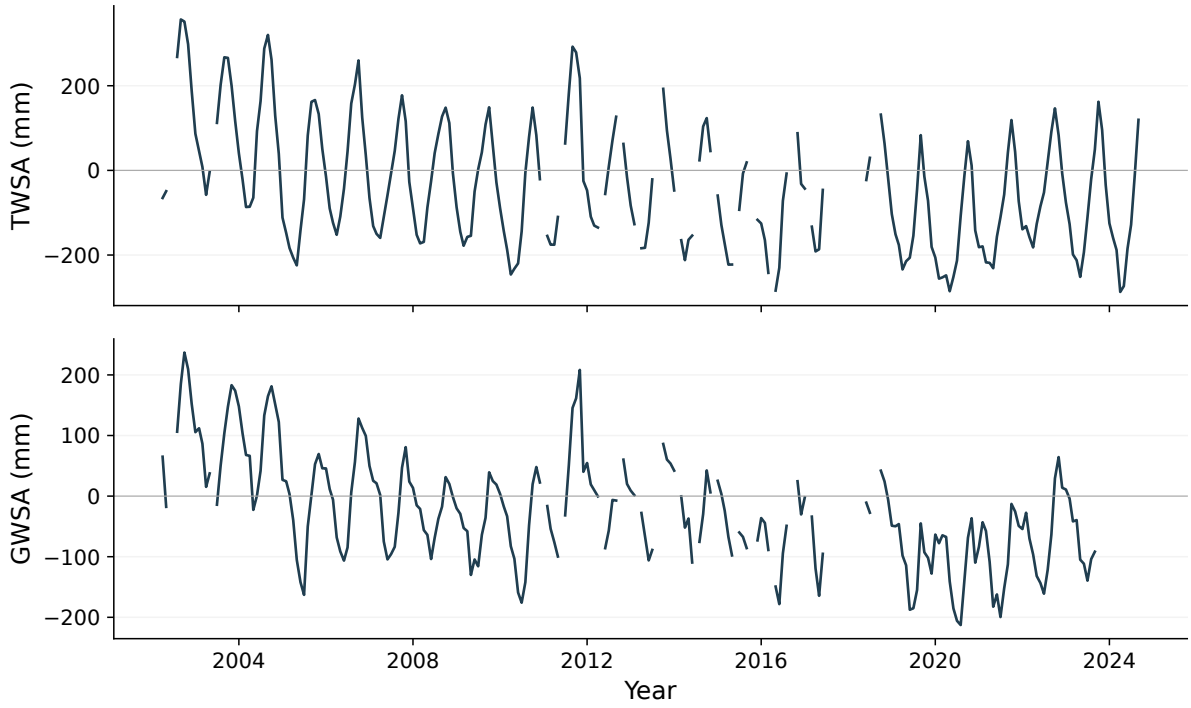
Step 6 compares annual residuals with groundwater levels. Depth to water is negated to give a head-oriented sign, and co-located wells are grouped before averaging. Raw, detrended and balanced-panel comparisons expose changing monitoring support. Head is not converted to storage without representative aquifer properties, so no cross-unit storage RMSE is reported.

Independent reservoir scenes check one surface-water component of the total-storage residual. The primary comparison uses a fixed pre-2004 reservoir population, requires scene support, removes season and trend, and resamples calendar years for a conditional correlation interval. Reservoirs respond to climate and operations together.

ENSO diagnostics use ONI and historical RONI v5, with DMI representing IOD. Selected leads are 0, 3, 6, 9 and 12 months. Partial correlations remove calendar month, linear trend and contemporaneous DMI. Twenty whole-year circular rotations of the 21-year calendar give a minimum diagnostic probability of 1/21; all candidate tests enter the FDR family.

ENSO phases require five consecutive overlapping seasons at $\pm 0.5°C$. Phase summaries are descriptive, and held-out model scores are calculated without refitting. No non-significant result proves complete climate removal. The uncertainty assessment keeps product choice, reconstruction error, discrepancy and monitoring coverage visible as separate limitations.

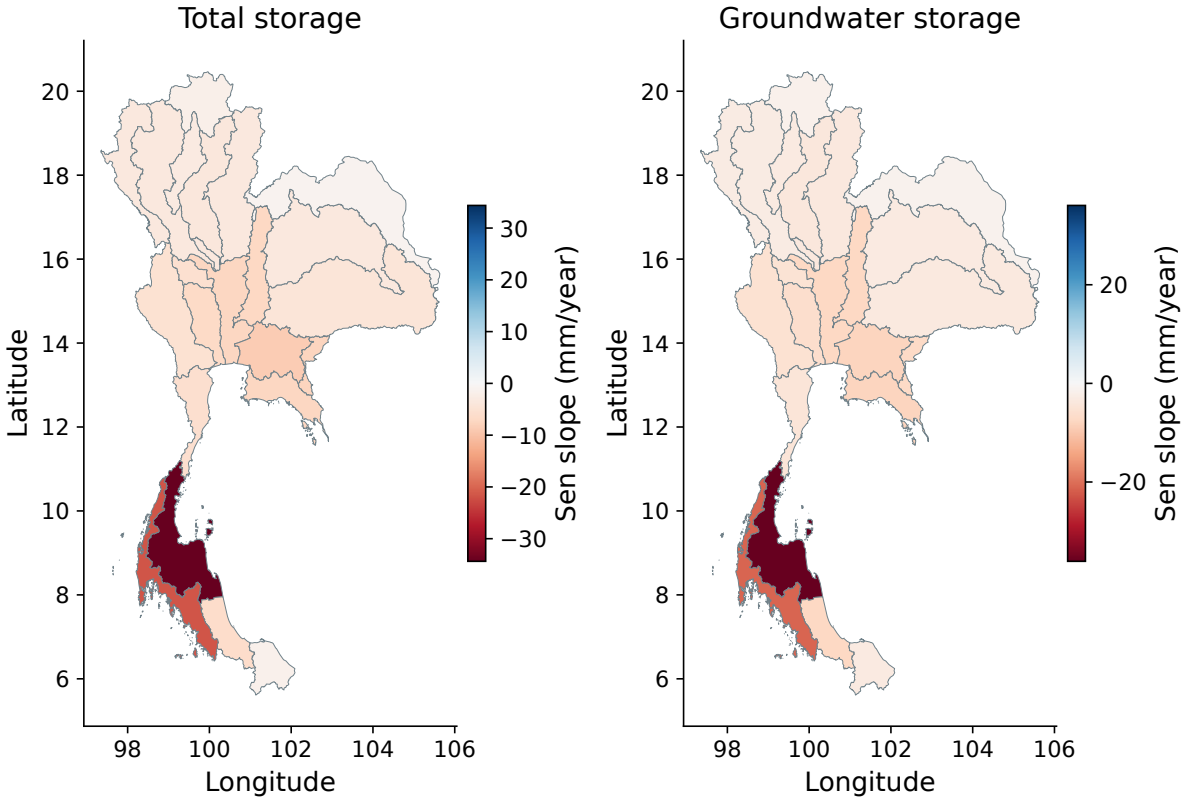
## 8 Results national storage change




**National total and groundwater storage**. TWSA uses the primary CSR record through September 2024; derived GWSA ends September 2023. Missing satellite months remain gaps. Negative anomalies mean storage below the 2004–2009 reference, not a negative absolute water volume.

**Interpretation.** National total and derived groundwater storage decline over their respective primary records. Seasonal Sen rates are −7.51 and −6.84 mm/year. The curves still rise during wet periods: a long-term decline does not imply a steady decrease every month.

## 8 Results spatial storage trends




**Total storage / Groundwater storage**. Seasonal Sen slopes for all 22 basins. Negative rates indicate decline. Basin significance, intervals and sample counts are reported separately. All basins share broad GRACE support.

**Interpretation.** The mapped rates differ among basins, but neighbouring polygons share gravity information. Strong southern rates warrant investigation alongside product sensitivity. Treat the map as regional screening rather than a precise ranking of independent aquifers.

## 8 Results trend values for all basins

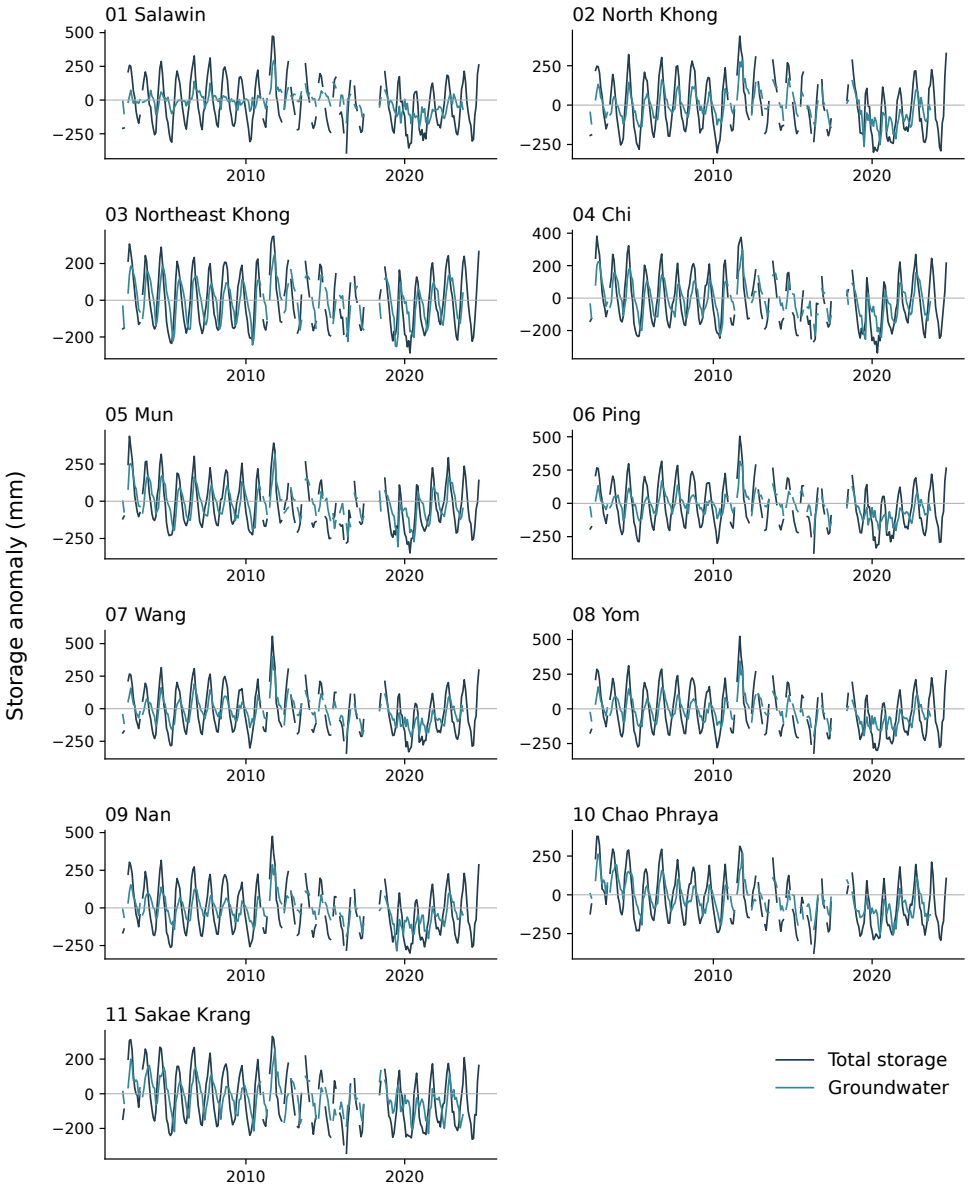
| Basin | TWS slope | TWS q | n | GWS slope | GWS q | n |
|---|---|---|---|---|---|---|
| 01 | -3.59 | 0.000 | 237 | -2.88 | 0.000 | 225 |
| 02 | -1.90 | 0.010 | 237 | -1.83 | 0.014 | 225 |
| 03 | -1.32 | 0.007 | 237 | -1.59 | 0.000 | 225 |
| 04 | -3.65 | 0.000 | 237 | -3.32 | 0.000 | 225 |
| 05 | -4.42 | 0.000 | 237 | -3.47 | 0.000 | 225 |
| 06 | -3.71 | 0.000 | 237 | -3.10 | 0.000 | 225 |
| 07 | -3.21 | 0.000 | 237 | -3.32 | 0.000 | 225 |
| 08 | -3.79 | 0.000 | 237 | -3.97 | 0.000 | 225 |
| 09 | -3.74 | 0.000 | 237 | -3.94 | 0.000 | 225 |
| 10 | -7.36 | 0.000 | 237 | -7.47 | 0.000 | 225 |
| 11 | -5.50 | 0.000 | 237 | -5.20 | 0.000 | 225 |
| 12 | -7.06 | 0.000 | 237 | -7.35 | 0.000 | 225 |
| 13 | -6.89 | 0.000 | 237 | -6.36 | 0.000 | 225 |
| 14 | -5.25 | 0.000 | 237 | -5.48 | 0.000 | 225 |
| 15 | -8.64 | 0.000 | 237 | -8.12 | 0.000 | 225 |
| 16 | -7.68 | 0.000 | 237 | -7.30 | 0.000 | 225 |
| 17 | -7.51 | 0.000 | 237 | -7.92 | 0.000 | 225 |
| 18 | -5.85 | 0.000 | 237 | -4.75 | 0.000 | 225 |
| 19 | -34.37 | 0.000 | 237 | -36.08 | 0.000 | 225 |
| 20 | -6.25 | 0.000 | 237 | -7.45 | 0.000 | 225 |
| 21 | -1.62 | 0.001 | 237 | -3.16 | 0.000 | 225 |
| 22 | -21.48 | 0.000 | 237 | -20.95 | 0.000 | 225 |

Slopes are seasonal Sen estimates in mm/year. q is seasonal Mann–Kendall FDR across basins; n is observed months. Full intervals and original p-values remain in CSR_basin_trends.csv. Basin names: see temporal panels.

Basin names: see temporal panels.

**Interpretation.** Read slope magnitude, adjusted probability and the number of observed months together. A small q value is evidence of ordered change within this test; it does not identify pumping or remove all serial-dependence uncertainty.

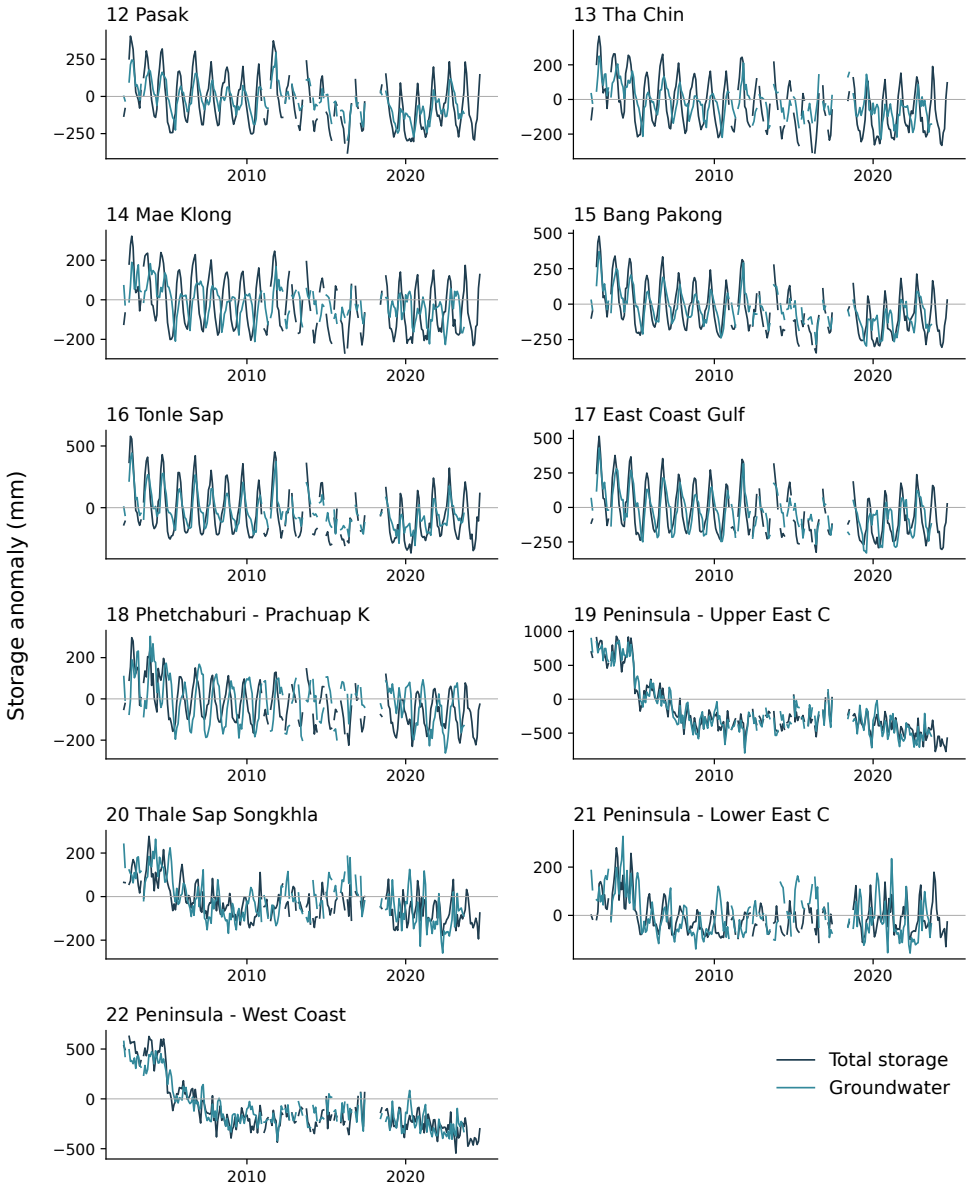
# 8 Results observed histories basins 01 to 11




**Observed histories for basins 01–11**. All basins are shown across two pages to keep labels legible. Total and derived groundwater anomalies have different component definitions and record endpoints.

**Interpretation.** The first eleven basin histories retain seasonality, interannual events and missing satellite months. Groundwater and total storage differ because the groundwater balance removes other compartments. Compare timing and persistence rather than assuming the two amplitudes represent the same quantity.

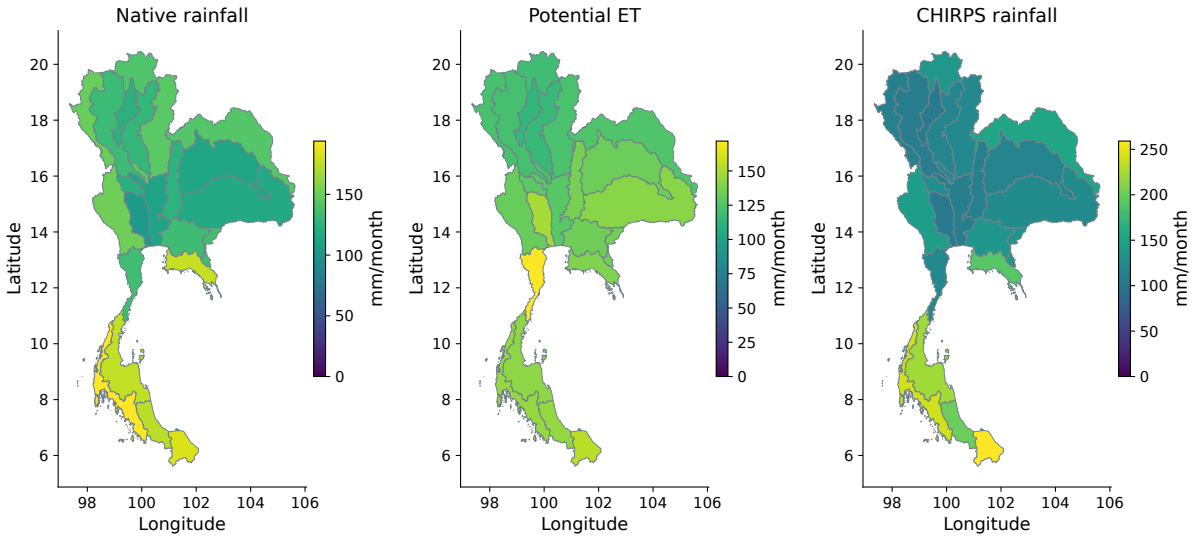
# 8 Results observed histories basins 12 to 22




**Observed histories for basins 12–22**. All basins are shown across two pages to keep labels legible. Total and derived groundwater anomalies have different component definitions and record endpoints.

**Interpretation.** The remaining basin histories show why a single national curve is insufficient. Different trajectories can coexist with national decline. Both the trend window and the component data endpoint influence a basin's displayed rate.

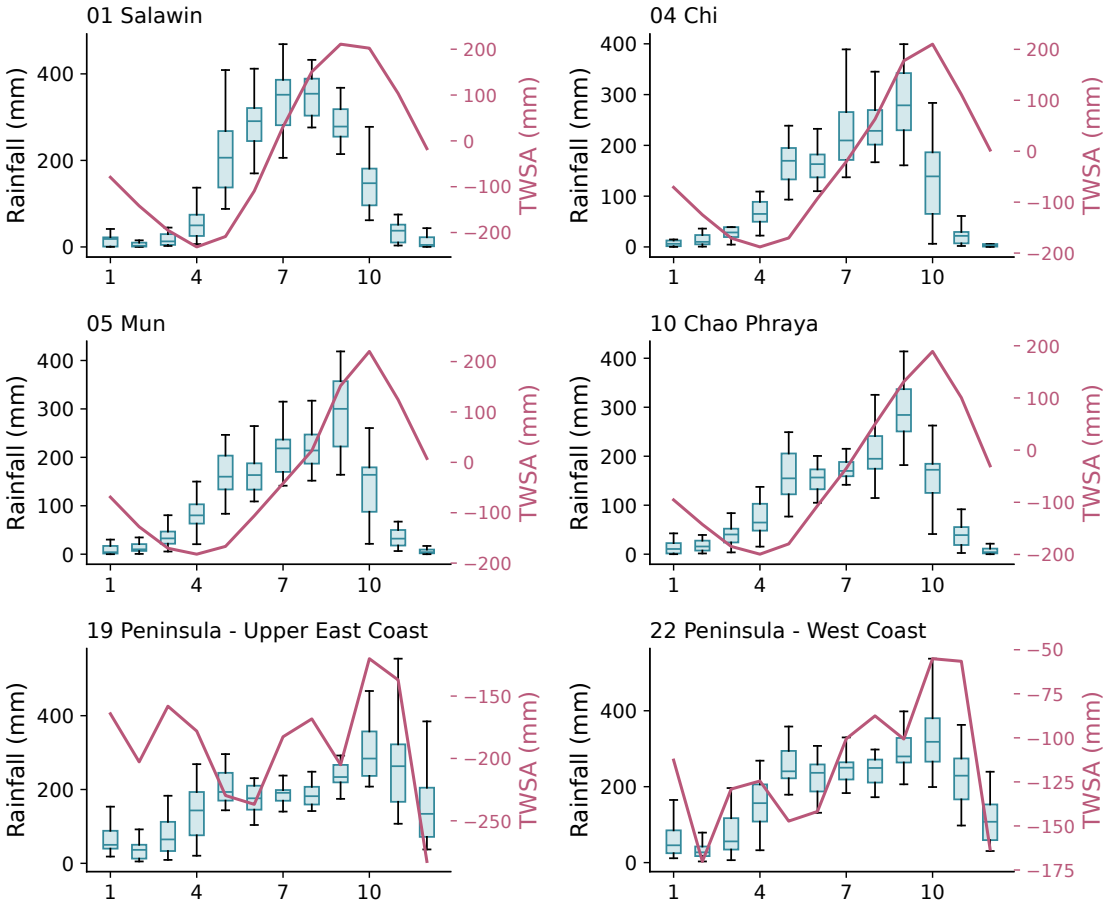
# 8 Results forcing patterns and processing sensitivity




**Native rainfall / Potential ET / CHIRPS rainfall**. Available-month means, 2003–2023. Native rainfall and PET drive the reconstruction; CHIRPS supplies alternative rainfall. Monthly fluxes are distinct from storage anomalies.

**Interpretation.** Rainfall and atmospheric demand vary spatially and supply physical context for storage. These are monthly fluxes, not stored water. Native and CHIRPS rainfall differ in estimation and forcing compatibility; neither is declared more accurate from parent-model skill alone.

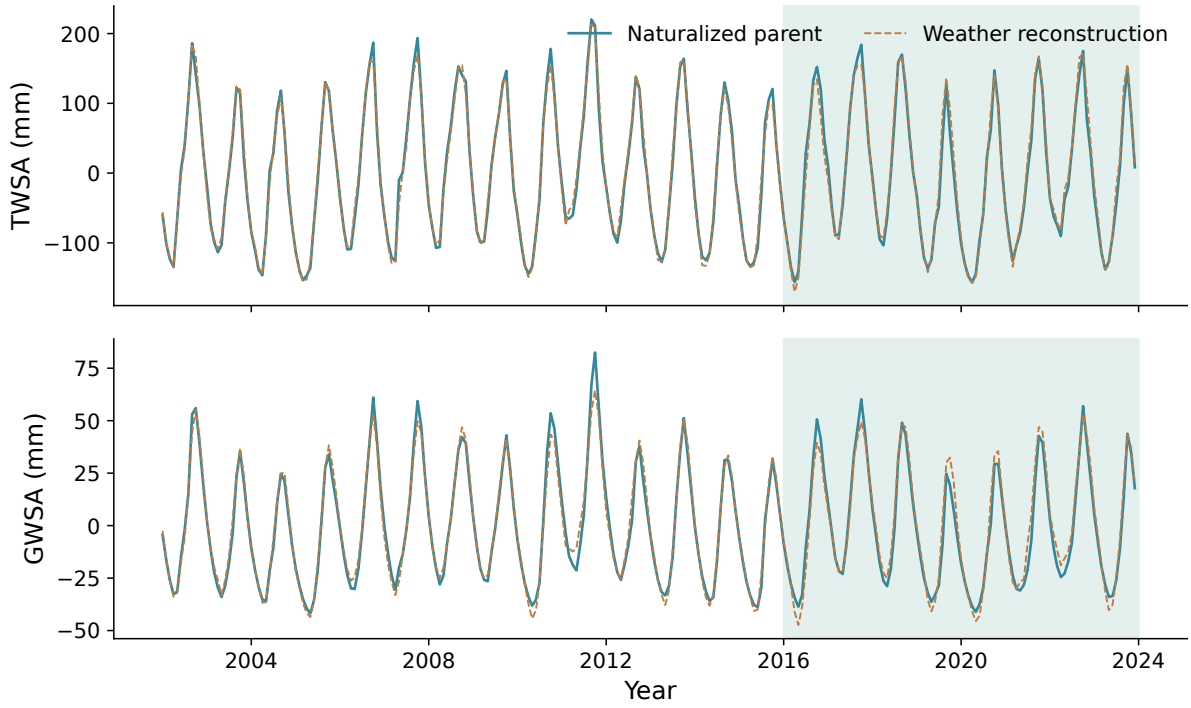
# 8 Results seasonal rainfall and storage




**Rainfall and storage seasonal timing**. Six geographically distributed illustrative basins; all 22 remain in the numerical release. Boxes show interannual native rainfall by month (quartiles and 1.5-IQR whiskers; outliers suppressed); pink line is mean observed TWSA on its own right-hand axis. Distribution spread is not a confidence interval.

**Interpretation.** Storage generally reflects accumulated hydrological inputs rather than rainfall in one month. The separate axes show rainfall distribution and mean storage timing. These six examples span geography; all 22 basin records remain in the tables and interactive site.

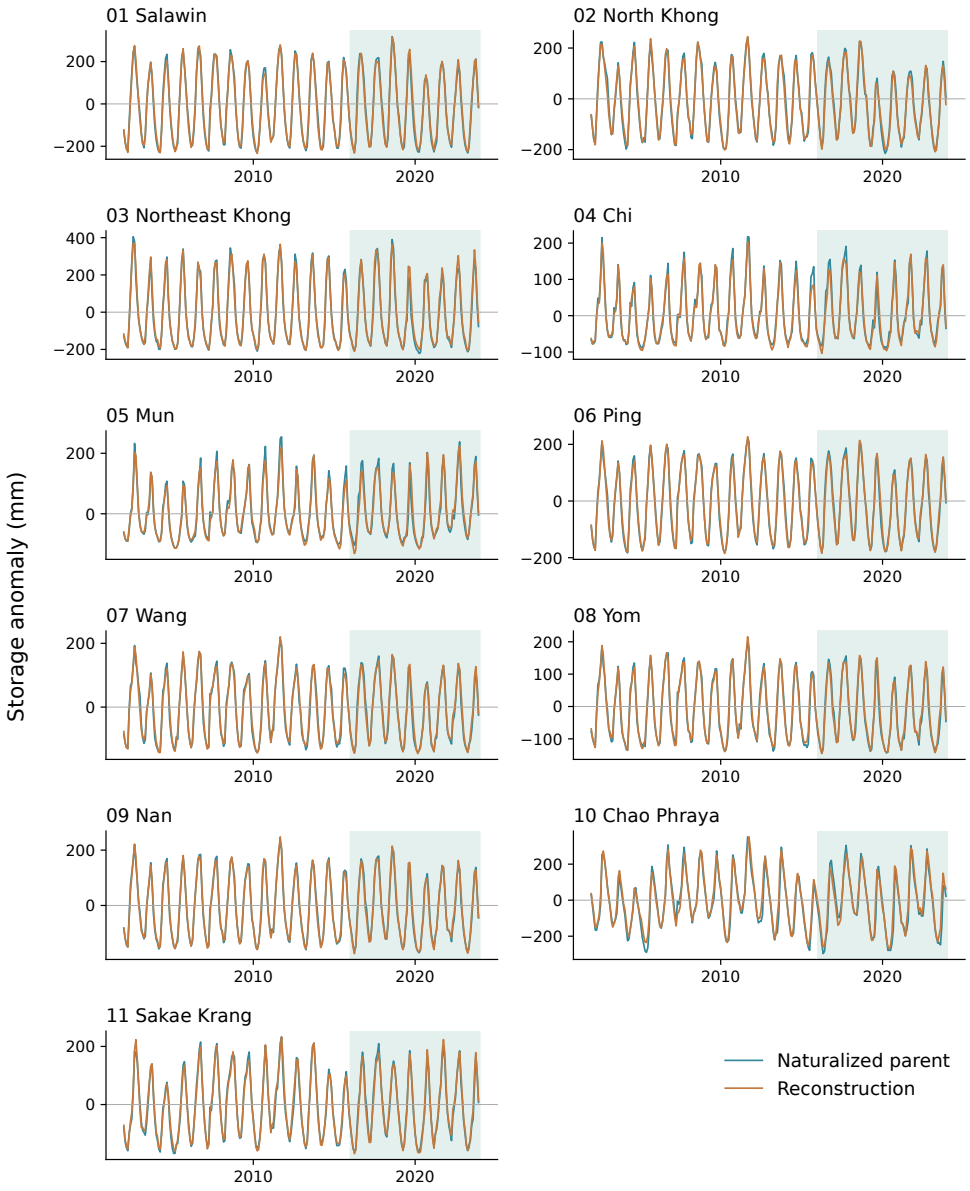
# 8 Results national climate reconstruction




**Weather reconstruction of naturalized storage**. The shaded years 2016–2023 were excluded from training and model selection. Close curves show reconstruction of the parent model; they do not independently establish the true natural climate counterfactual.

**Interpretation.** The weather model follows the naturalized parent closely in the shaded test years. National held-out NSE is 0.981 for total storage and 0.961 for groundwater. This is strong reconstruction of the parent, not independent validation of the natural counterfactual.

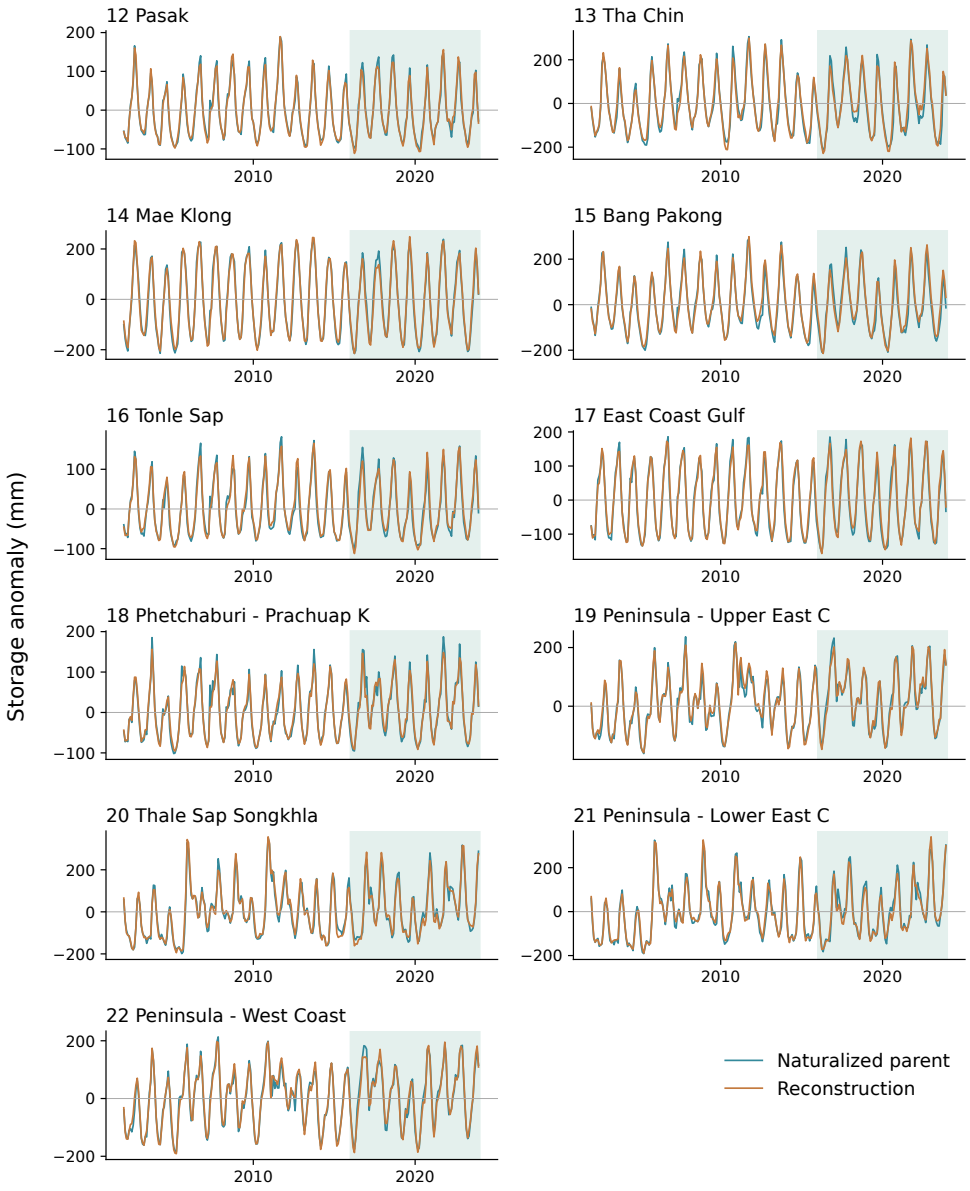
# 8 Results total storage reconstruction basins 01 to 11




**Climate histories for basins 01–11**. All basins are shown across two pages to keep labels legible. Shading marks the held-out test years 2016–2023. The parent is naturalized WaterGAP, not observed climate-only storage.

**Interpretation.** The shaded test period was excluded from fitting and model selection. Close curves indicate the emulator reproduces modeled naturalized total storage. Earlier training and validation context must not be counted as independent test evidence.

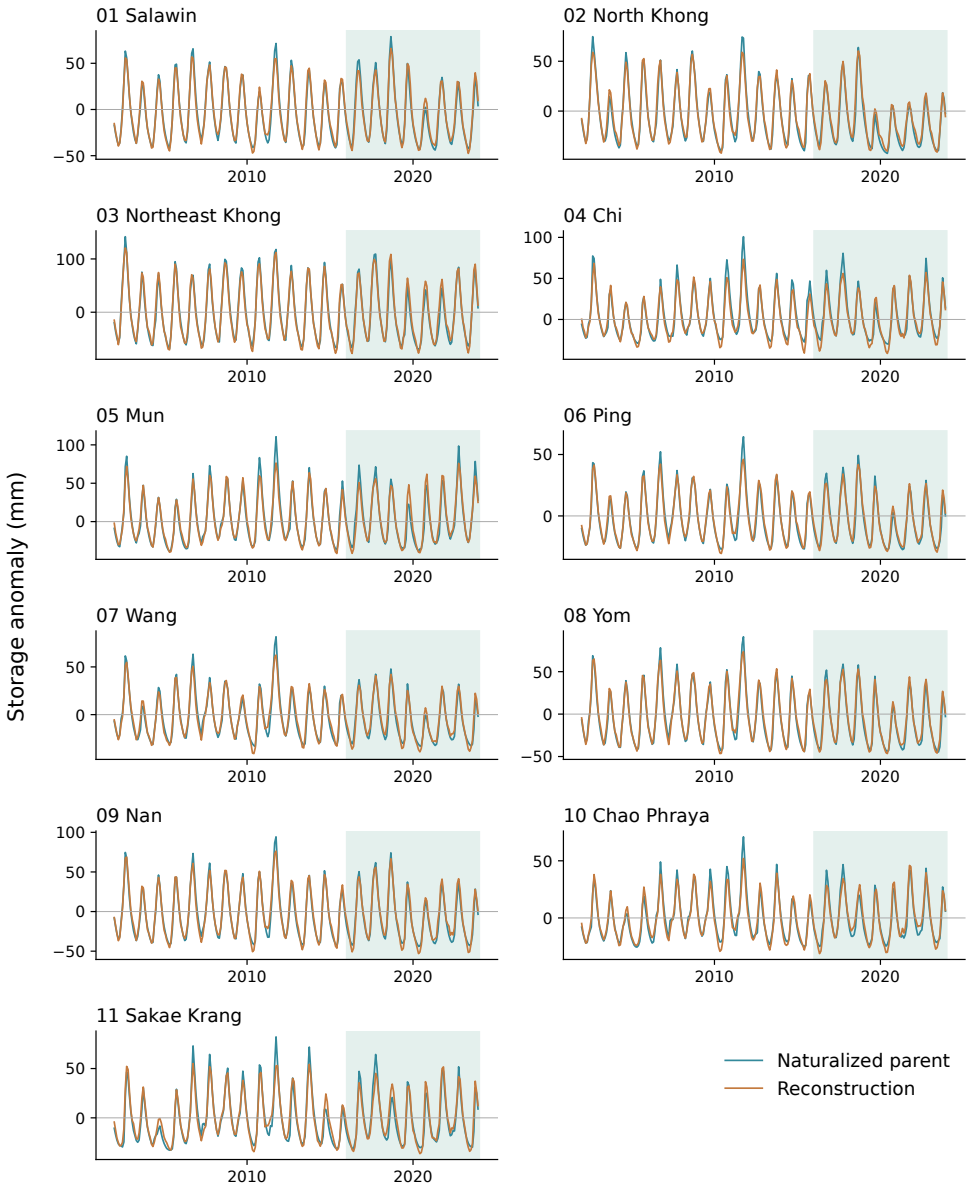
# 8 Results total storage reconstruction basins 12 to 22




**Climate histories for basins 12–22**. All basins are shown across two pages to keep labels legible. Shading marks the held-out test years 2016–2023. The parent is naturalized WaterGAP, not observed climate-only storage.

**Interpretation.** All basins are retained, including differences in amplitude and timing. The model target uses naturalized storage; the agreement shown here does not establish the cause of any GRACE-model mismatch.

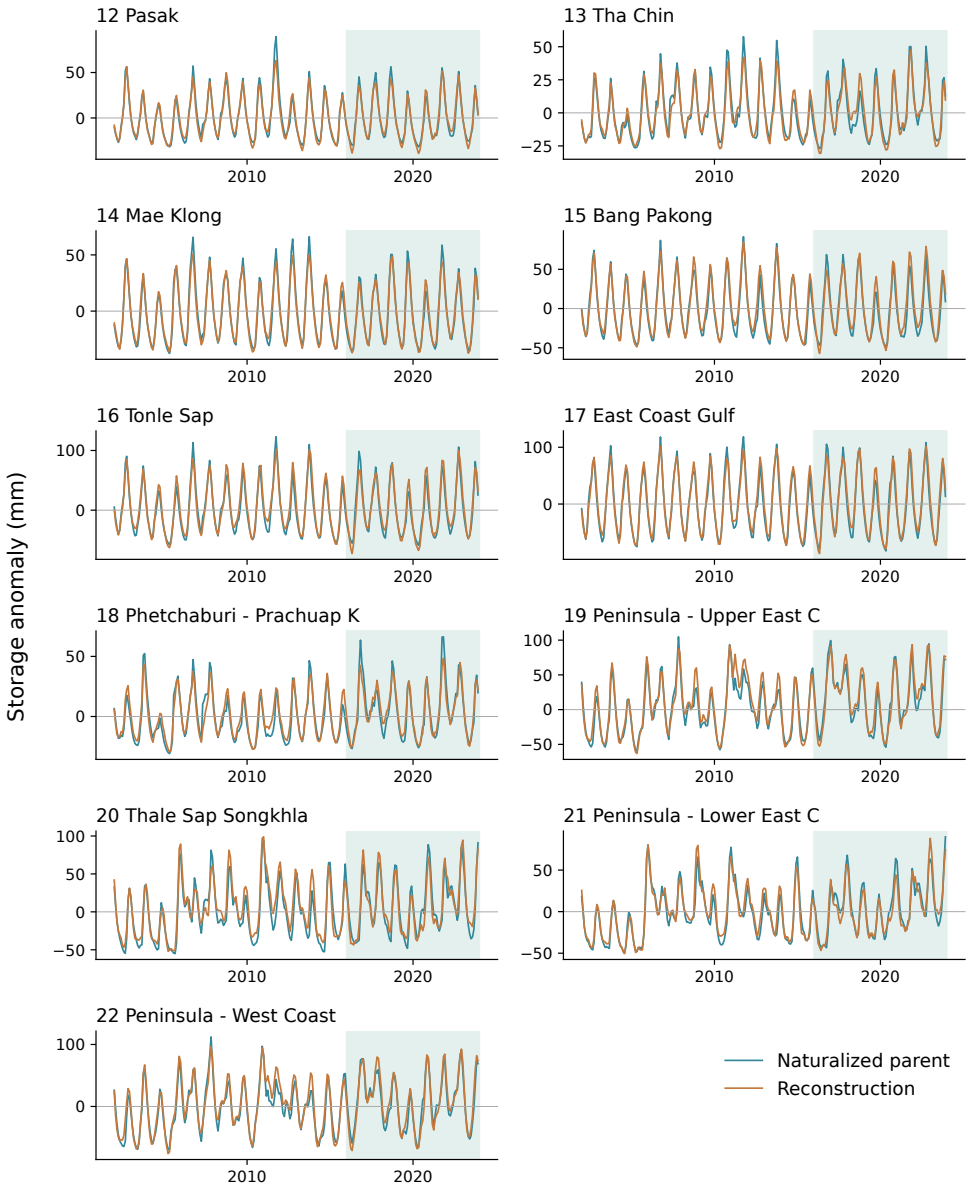
# 8 Results groundwater reconstruction basins 01 to 11




**Climate histories for basins 01–11**. All basins are shown across two pages to keep labels legible. Shading marks the held-out test years 2016–2023. The parent is naturalized WaterGAP, not observed climate-only storage.

**Interpretation.** Groundwater response is modeled through meteorological memory and parent hydrology. Good agreement shows that weather history reconstructs the parent signal. It does not validate pumping, specific yield or measured aquifer storage.

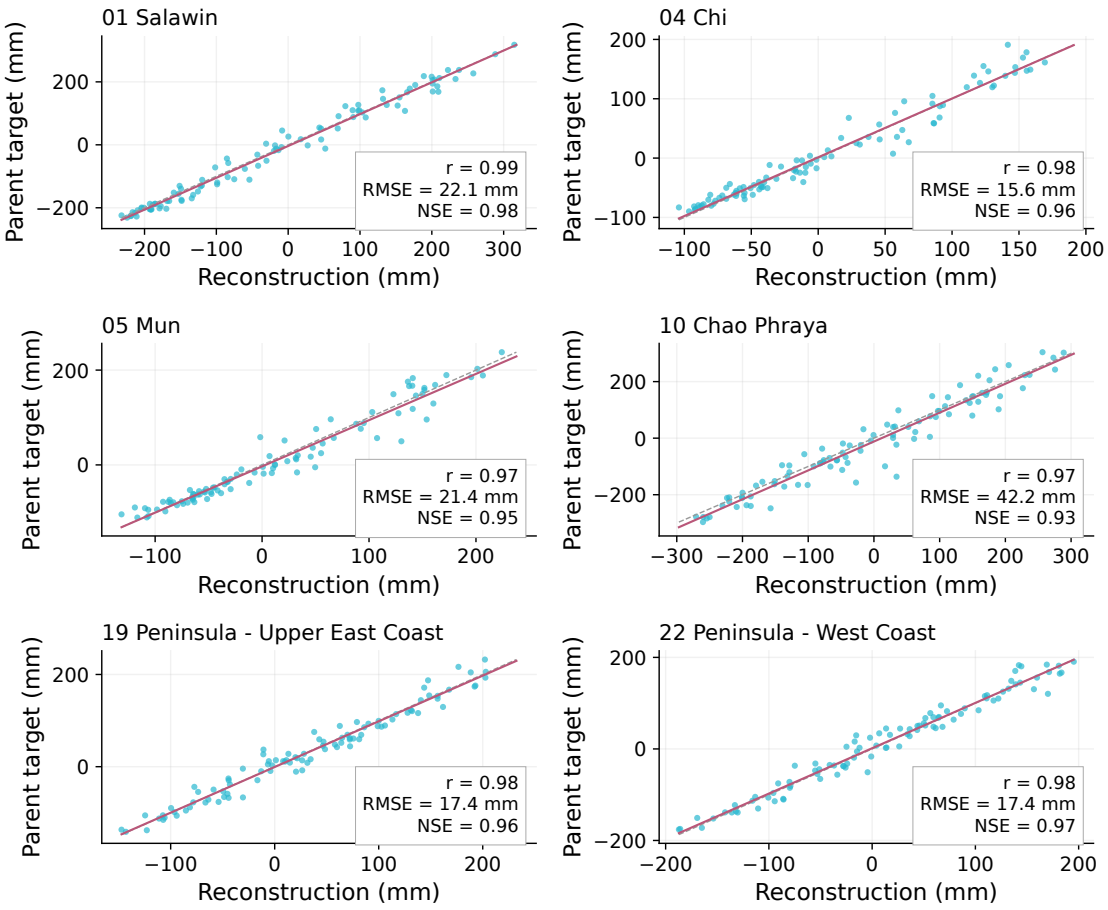
# 8 Results groundwater reconstruction basins 12 to 22




**Climate histories for basins 12–22**. All basins are shown across two pages to keep labels legible. Shading marks the held-out test years 2016–2023. The parent is naturalized WaterGAP, not observed climate-only storage.

**Interpretation.** The second half of the groundwater reconstruction retains the same frozen test design. Basin-level metrics and annual failures are needed alongside these visually close curves to judge performance fairly.

## 8 Results held out model agreement




**Held-out TWSA reconstruction agreement**. 2016–2023 test months, six geographical examples. Grey is identity; pink is a descriptive fit. Metrics compare predictions directly with the parent target. Basin selection is geographic, not based on model performance; all 22 basin metrics are provided.

**Interpretation.** Points near the grey identity line indicate small reconstruction errors. Correlation, RMSE and NSE evaluate the predictions directly. The pink descriptive fit cannot improve those reported scores. Geographical examples are shown here; the next tables retain every basin.

## 8 Results total storage model scores

| Basin | n | r | NSE | Anom NSE | RMSE | MAE |
|---|---|---|---|---|---|---|
| 01 | 96 | 0.991 | 0.981 | 0.807 | 22.08 | 16.90 |
| 02 | 96 | 0.990 | 0.979 | 0.904 | 17.32 | 13.88 |
| 03 | 96 | 0.990 | 0.973 | 0.812 | 27.37 | 19.39 |
| 04 | 96 | 0.981 | 0.961 | 0.821 | 15.61 | 11.58 |
| 05 | 96 | 0.974 | 0.946 | 0.732 | 21.43 | 15.40 |
| 06 | 96 | 0.991 | 0.981 | 0.876 | 15.90 | 12.00 |
| 07 | 96 | 0.987 | 0.974 | 0.868 | 15.24 | 11.08 |
| 08 | 96 | 0.987 | 0.972 | 0.873 | 15.38 | 11.43 |
| 09 | 96 | 0.992 | 0.983 | 0.901 | 14.14 | 10.83 |
| 10 | 96 | 0.967 | 0.928 | 0.792 | 42.23 | 30.13 |
| 11 | 96 | 0.990 | 0.980 | 0.875 | 15.34 | 12.27 |
| 12 | 96 | 0.987 | 0.974 | 0.864 | 11.22 | 8.35 |
| 13 | 96 | 0.974 | 0.942 | 0.778 | 32.00 | 23.99 |
| 14 | 96 | 0.991 | 0.981 | 0.856 | 18.49 | 14.13 |
| 15 | 96 | 0.975 | 0.942 | 0.764 | 28.65 | 22.72 |
| 16 | 96 | 0.979 | 0.957 | 0.705 | 14.60 | 11.45 |
| 17 | 96 | 0.987 | 0.972 | 0.742 | 16.75 | 13.62 |
| 18 | 96 | 0.979 | 0.957 | 0.827 | 14.38 | 10.97 |
| 19 | 96 | 0.982 | 0.965 | 0.911 | 17.38 | 13.55 |
| 20 | 96 | 0.982 | 0.963 | 0.903 | 22.54 | 18.01 |
| 21 | 96 | 0.977 | 0.951 | 0.887 | 25.44 | 18.76 |
| 22 | 96 | 0.985 | 0.970 | 0.903 | 17.36 | 13.16 |

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

**Interpretation.** Anomaly NSE asks whether the model reproduces departures from the usual seasonal cycle. It is more demanding than raw seasonal agreement. All scores use the same held-out 2016–2023 parent-model target, making within-table basin comparisons meaningful.

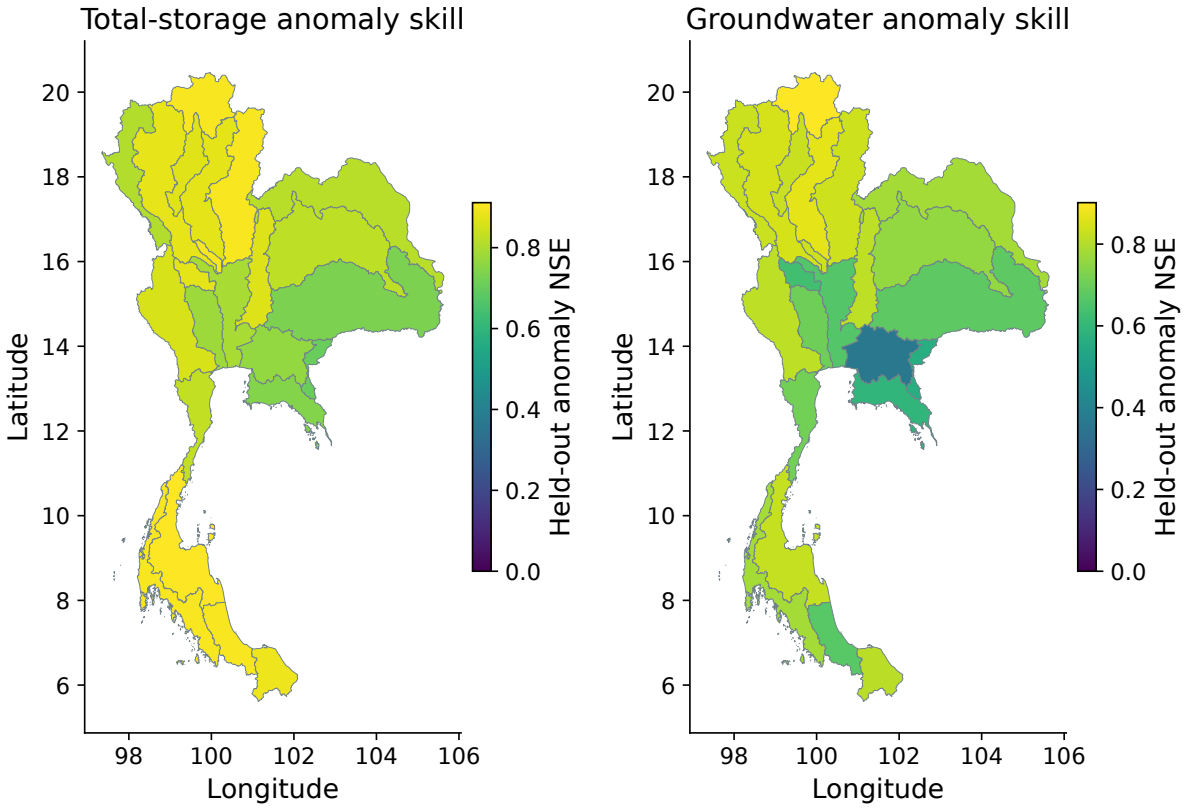
## 8 Results groundwater model scores

| Basin | n | r | NSE | Anom NSE | RMSE | MAE |
|---|---|---|---|---|---|---|
| 01 | 96 | 0.982 | 0.963 | 0.830 | 5.74 | 4.47 |
| 02 | 96 | 0.986 | 0.956 | 0.901 | 4.98 | 4.08 |
| 03 | 96 | 0.982 | 0.953 | 0.781 | 10.59 | 7.70 |
| 04 | 96 | 0.969 | 0.933 | 0.755 | 7.04 | 5.33 |
| 05 | 96 | 0.963 | 0.927 | 0.678 | 8.87 | 6.32 |
| 06 | 96 | 0.980 | 0.958 | 0.842 | 3.94 | 2.92 |
| 07 | 96 | 0.977 | 0.948 | 0.824 | 4.83 | 3.96 |
| 08 | 96 | 0.984 | 0.963 | 0.865 | 5.52 | 4.22 |
| 09 | 96 | 0.982 | 0.958 | 0.832 | 6.04 | 4.78 |
| 10 | 96 | 0.968 | 0.935 | 0.661 | 4.90 | 4.09 |
| 11 | 96 | 0.958 | 0.914 | 0.627 | 7.09 | 5.40 |
| 12 | 96 | 0.981 | 0.962 | 0.812 | 4.62 | 3.61 |
| 13 | 96 | 0.959 | 0.920 | 0.703 | 5.50 | 4.45 |
| 14 | 96 | 0.985 | 0.970 | 0.812 | 4.39 | 3.12 |
| 15 | 96 | 0.963 | 0.902 | 0.360 | 10.51 | 8.46 |
| 16 | 96 | 0.970 | 0.933 | 0.564 | 10.88 | 8.87 |
| 17 | 96 | 0.980 | 0.952 | 0.589 | 11.67 | 9.60 |
| 18 | 96 | 0.958 | 0.907 | 0.711 | 6.84 | 5.11 |
| 19 | 96 | 0.968 | 0.926 | 0.823 | 10.40 | 8.46 |
| 20 | 96 | 0.946 | 0.892 | 0.668 | 11.69 | 9.70 |
| 21 | 96 | 0.953 | 0.904 | 0.808 | 9.19 | 7.33 |
| 22 | 96 | 0.967 | 0.922 | 0.780 | 11.32 | 8.94 |

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

**Interpretation.** Groundwater errors are smaller in mm partly because the target has a different variance and component definition. Do not rank groundwater and total-storage models solely by their raw RMSE. Inspect anomaly skill within the same storage target.

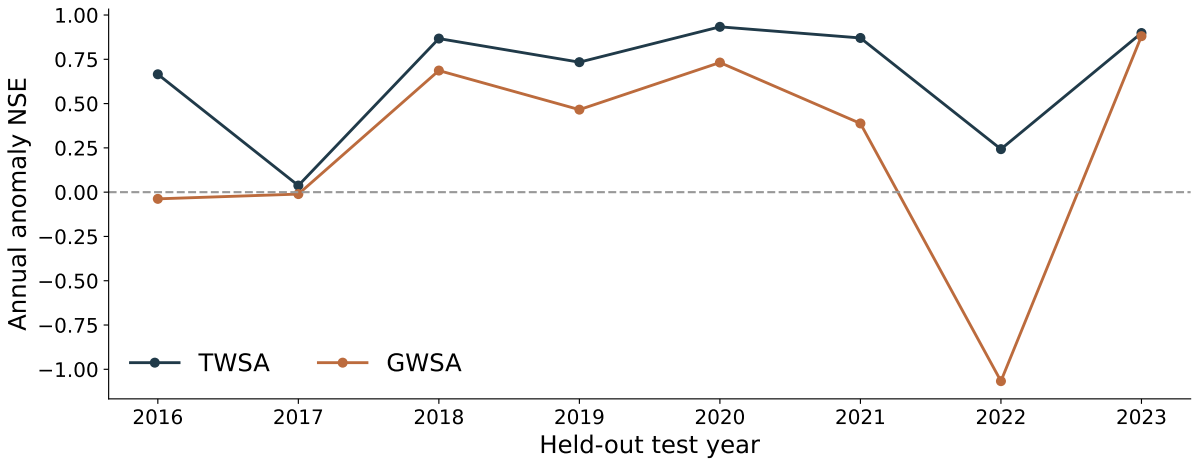
## 8 Results model performance in space




**Total-storage anomaly skill / Groundwater anomaly skill**. NSE compares squared error with the target-mean benchmark; 1 is perfect, 0 equals that benchmark and negative is worse. Monthly climatology comes only from training. The target is naturalized WaterGAP.

**Interpretation.** Anomaly skill differs among basins despite high pooled scores. The map identifies where parent-model reconstruction is less reliable. It remains a model evaluation map, not a map of confidence in human attribution.

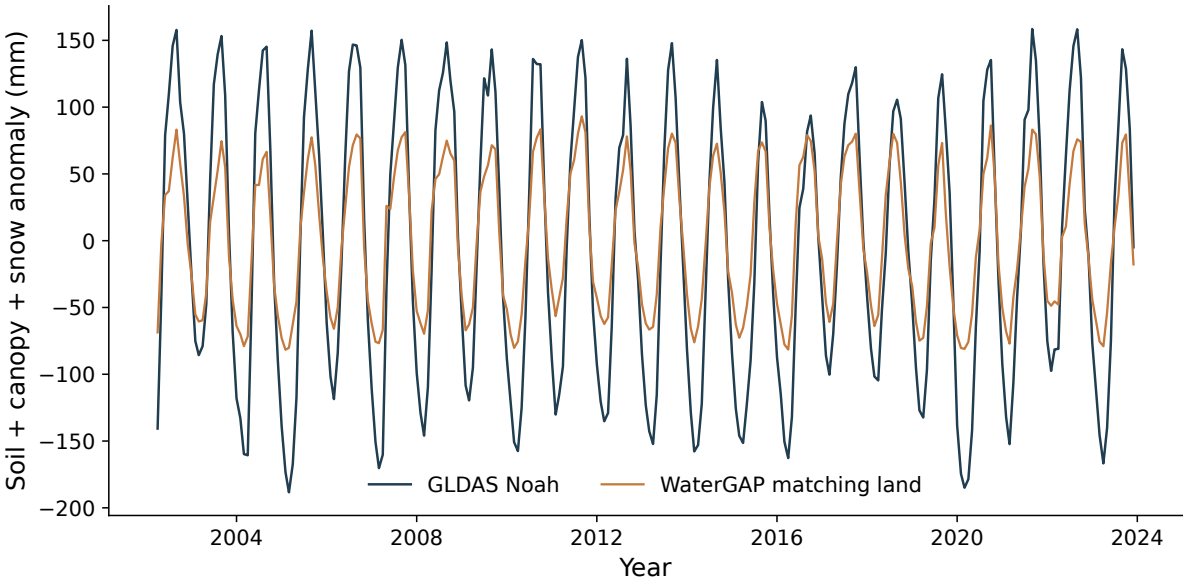
## 8 Results annual skill and climate memory




**Skill changes between test years**. National annual anomaly NSE against the naturalized parent. The 2022 groundwater score is −1.067 and is retained; pooled scores hide this failure.

**Interpretation.** The 2022 groundwater anomaly NSE is −1.067: in that year, the reconstruction is worse than its mean benchmark. The failure is retained. Separately, the controlled memory experiment improves national anomaly NSE from 0.283 to 0.890 for total storage and 0.039 to 0.723 for groundwater.

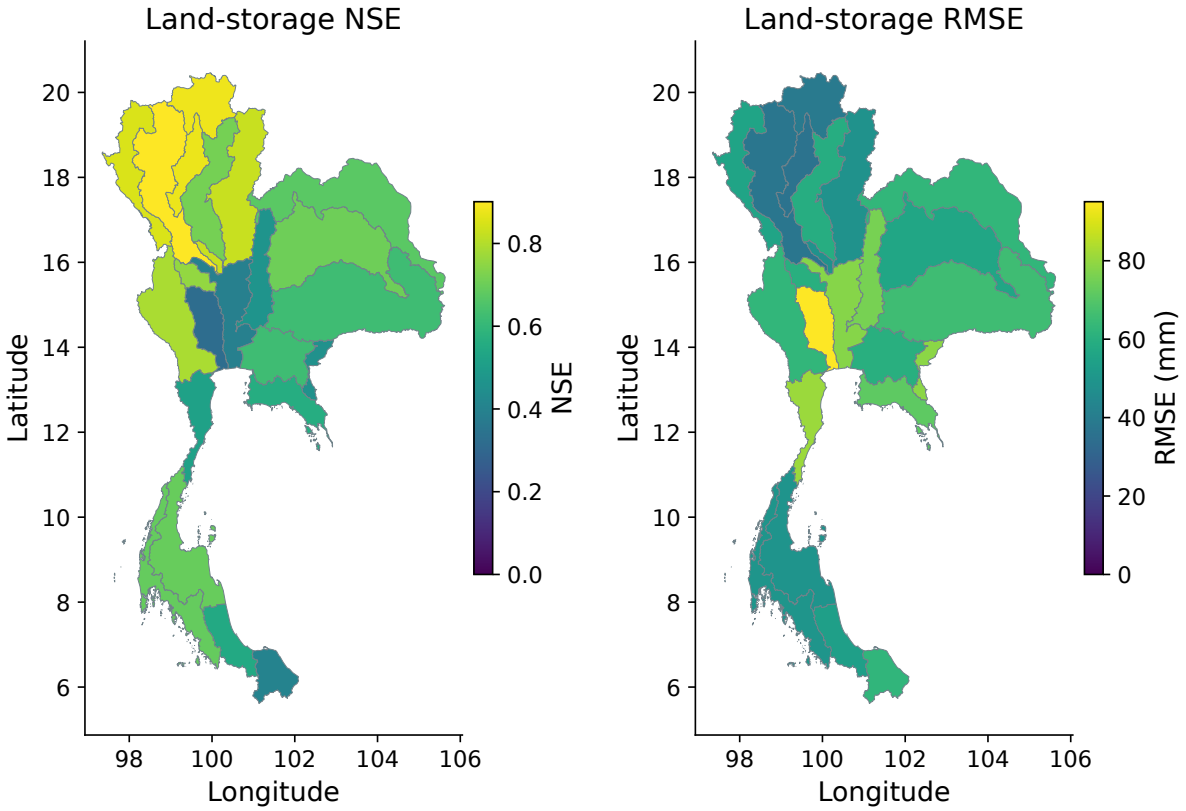
# 8 Results GLDAS temporal comparison




**Matching land-storage histories**. This comparison excludes groundwater and open surface water. GLDAS is a cross-model benchmark, not an observed total-storage or human-only validation dataset.

**Interpretation.** National matched land storage has r = 0.969, NSE = 0.738 and RMSE = 52.49 mm across 261 paired months. Similar timing supports land-component consistency. The amplitude differences matter, and this comparison excludes groundwater and surface water.

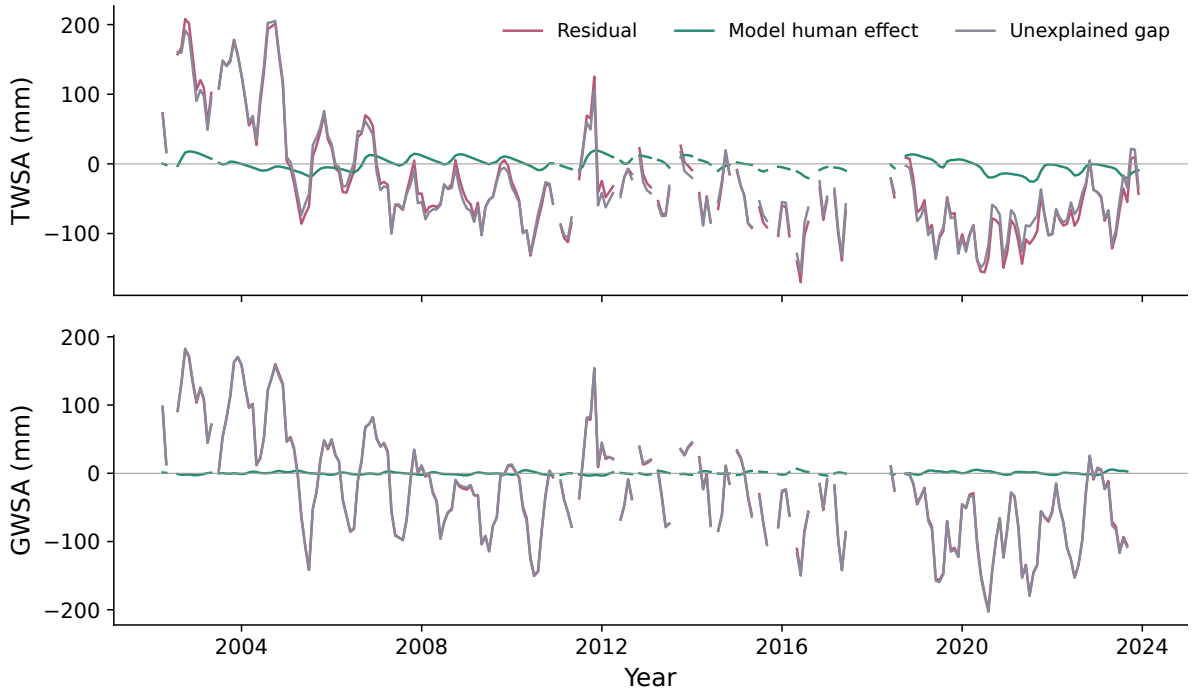
## 8 Results GLDAS basin scores




**Land-storage NSE / Land-storage RMSE**. Matching soil, canopy and snow compartments on paired dates. Higher NSE and lower RMSE mean closer model agreement for this same target. They do not validate the residual.

**Interpretation.** Higher NSE and lower RMSE indicate closer agreement for matching land compartments. They cannot validate total storage or human water use. Shared climate drivers and GLDAS involvement in groundwater subtraction limit independence.

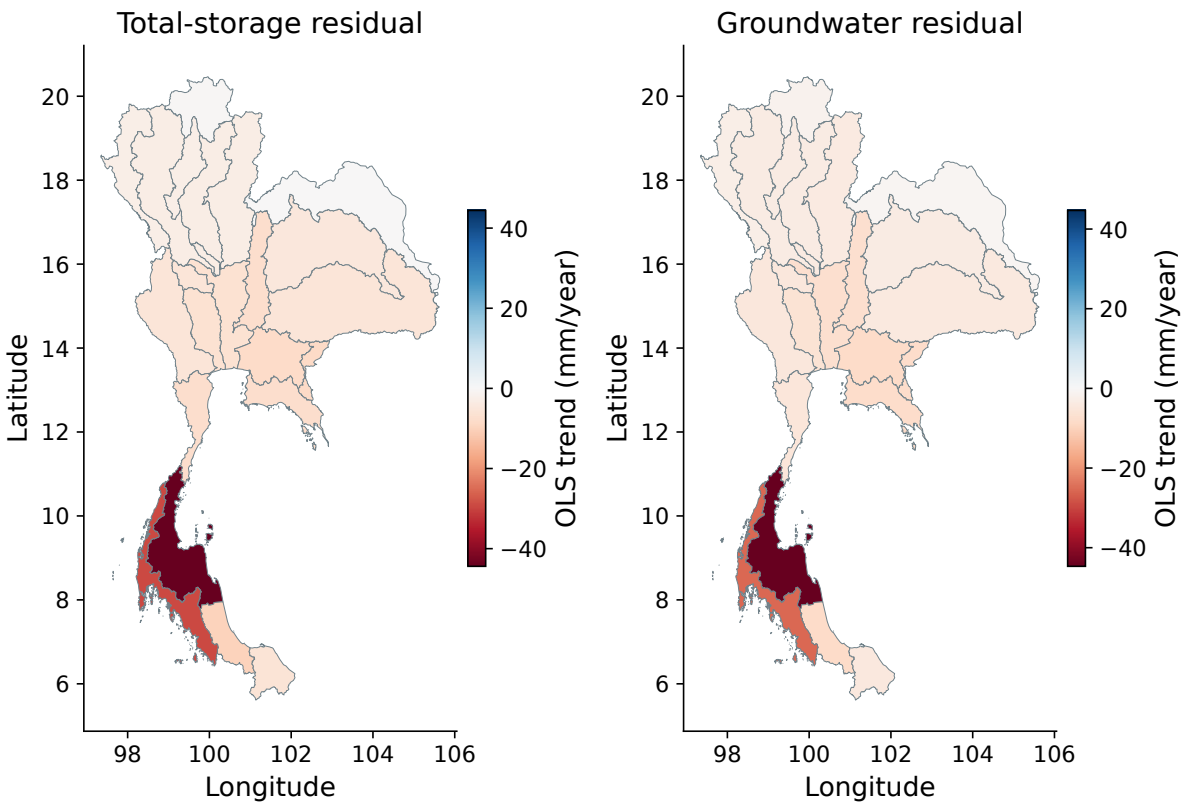
## 8 Results residual and represented human effect




**Residual compared with the modeled human effect**. Residual = observed − naturalized. Modeled human effect = historical − naturalized. Unexplained gap = observed − historical. Thus residual = modeled human effect + unexplained gap. A negative residual is not measured pumping.

**Interpretation.** The residual is much larger than the direct human effect represented by the paired model. Much of their difference remains an unexplained observation-model gap. The data therefore do not support labelling the entire residual as measured human depletion.

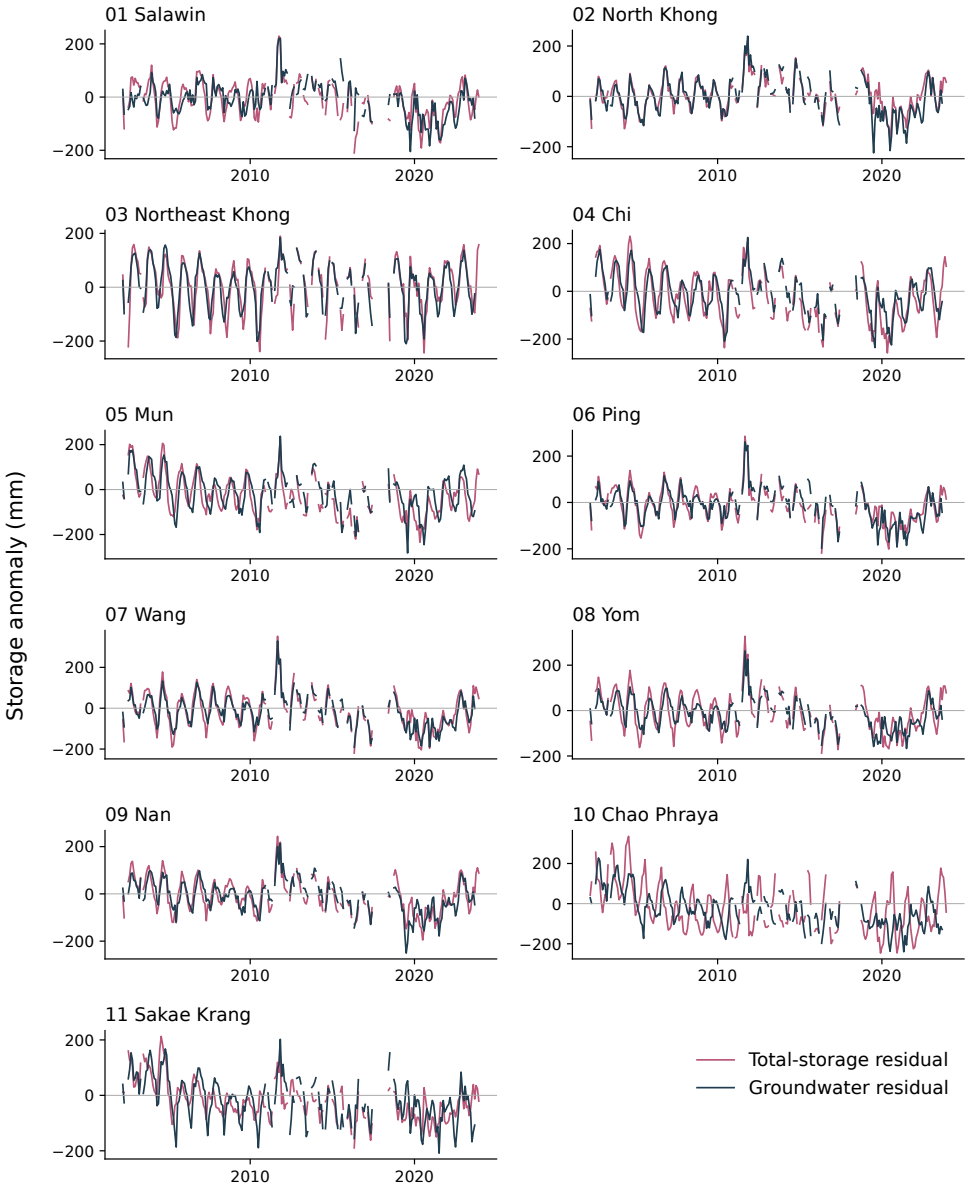
## 8 Results residual trend patterns




**Total-storage residual / Groundwater residual**. Calendar-month fixed-effect OLS on common available dates. These additive rates differ from the primary seasonal Sen estimates. Residuals include human effects and observation/model discrepancy.

**Interpretation.** Negative residual trends indicate observed storage falling relative to the naturalized baseline. They can combine human influence with observation and model discrepancy. OLS is used here so the additive accounting remains valid on the same months.

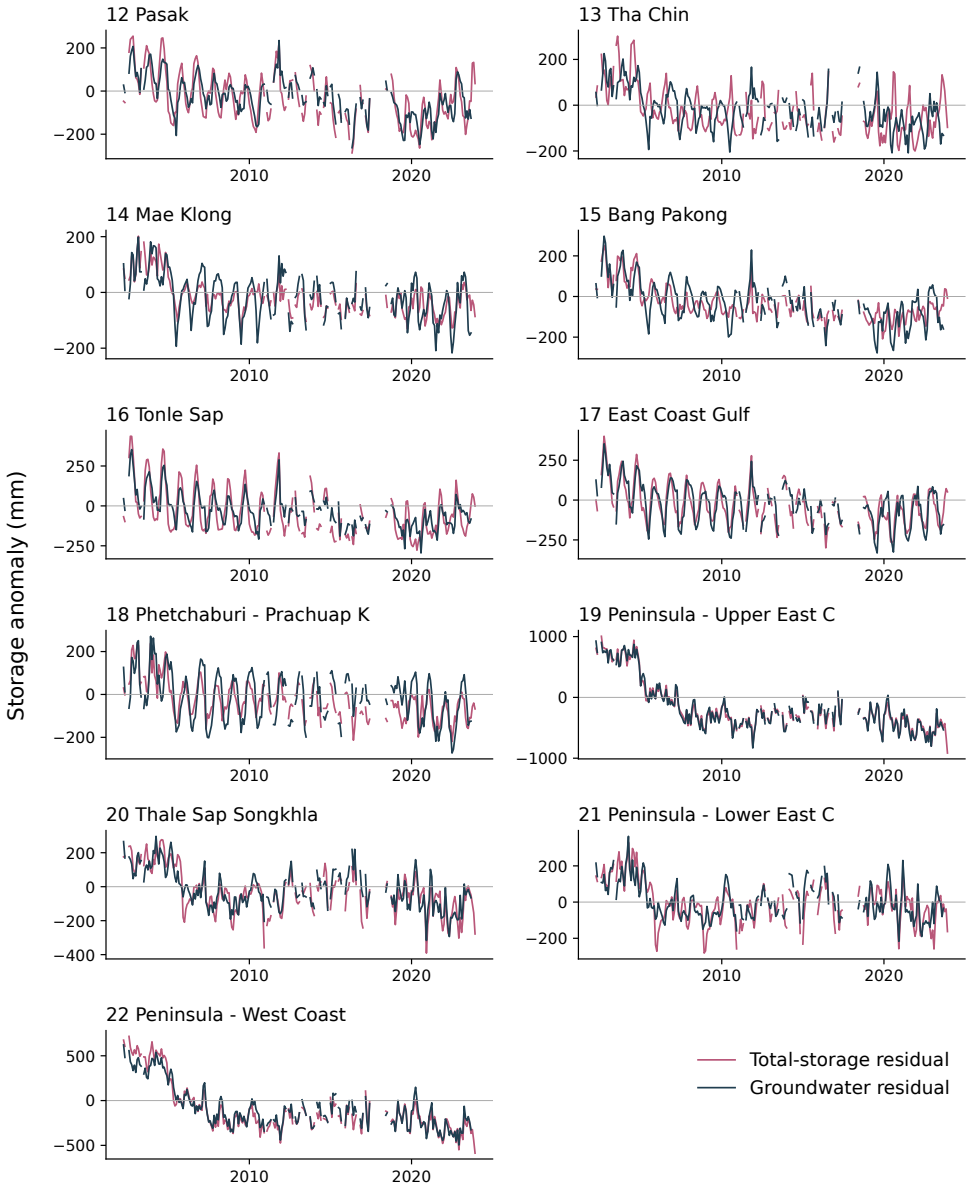
# 8 Results residual histories basins 01 to 11




**Residual histories for basins 01–11**. All basins are shown across two pages to keep labels legible. Real observation gaps remain missing. The residual is observed minus the naturalized baseline and includes unresolved discrepancy.

**Interpretation.** These histories show the total and groundwater residual on the same basin calendar. Persistence and events can be compared, but a negative value does not identify a unique cause. Missing observations remain gaps rather than inferred depletion.

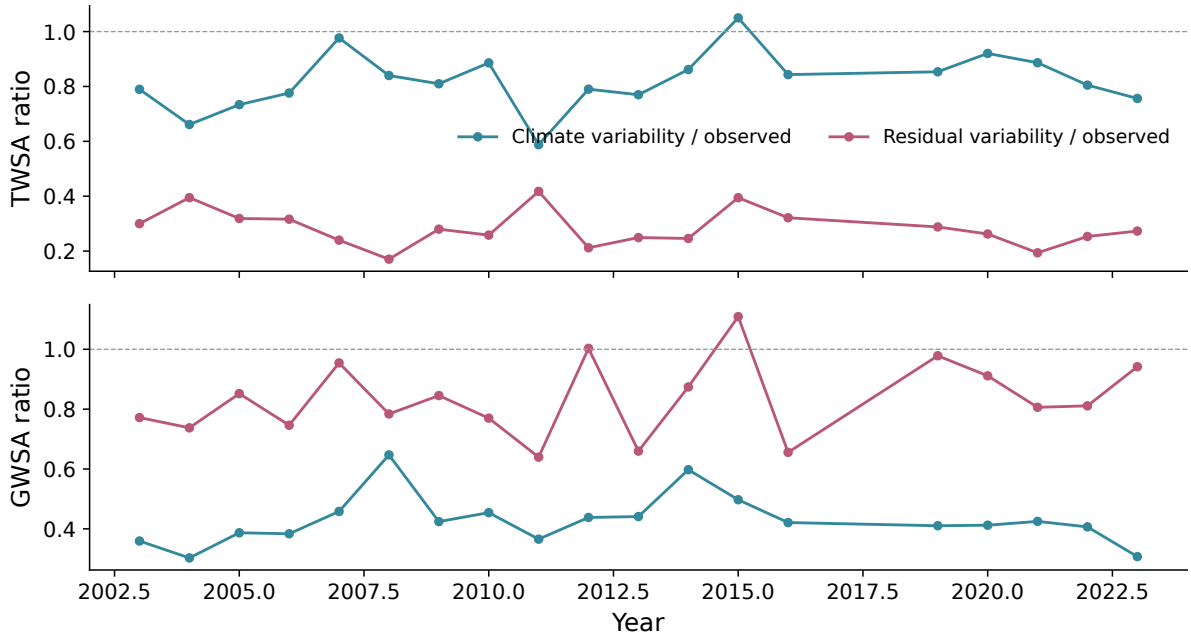
# 8 Results residual histories basins 12 to 22




**Residual histories for basins 12–22**. All basins are shown across two pages to keep labels legible. Real observation gaps remain missing. The residual is observed minus the naturalized baseline and includes unresolved discrepancy.

**Interpretation.** The remaining basins show that residual timing and magnitude vary across Thailand. Interpreting a basin requires the independent evidence and support shown later, not only the colour or sign of its trend.

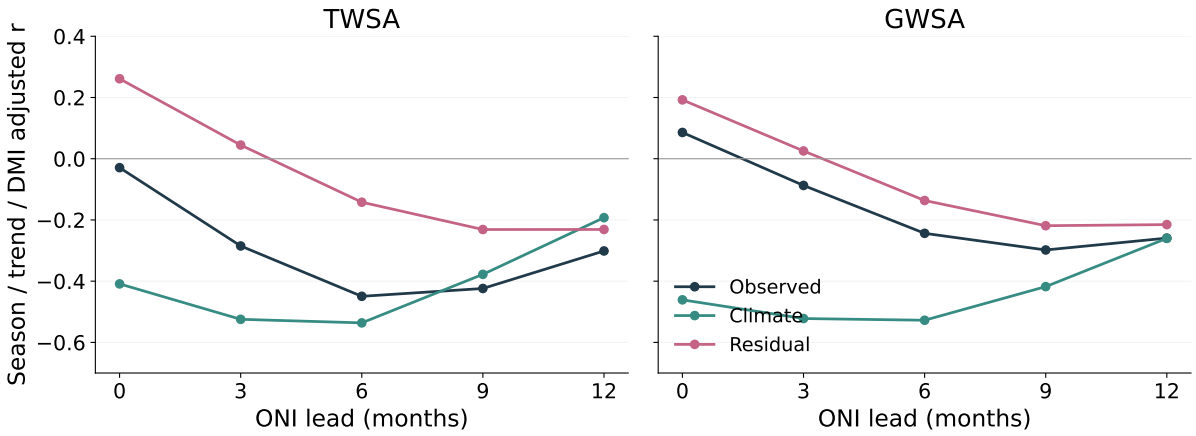
# 8 Results climate and residual variability




**How climate and residual variability change over time**. Annual mean absolute deviations on the same available months (minimum eight). Ratios need not sum to one and may exceed one because components covary. These are not climate/human percentages.

**Interpretation.** Climate and residual contributions to variability change through time. The ratios are based on mean absolute deviation and need not add to one. They must not be converted into complementary climate-versus-human percentages.

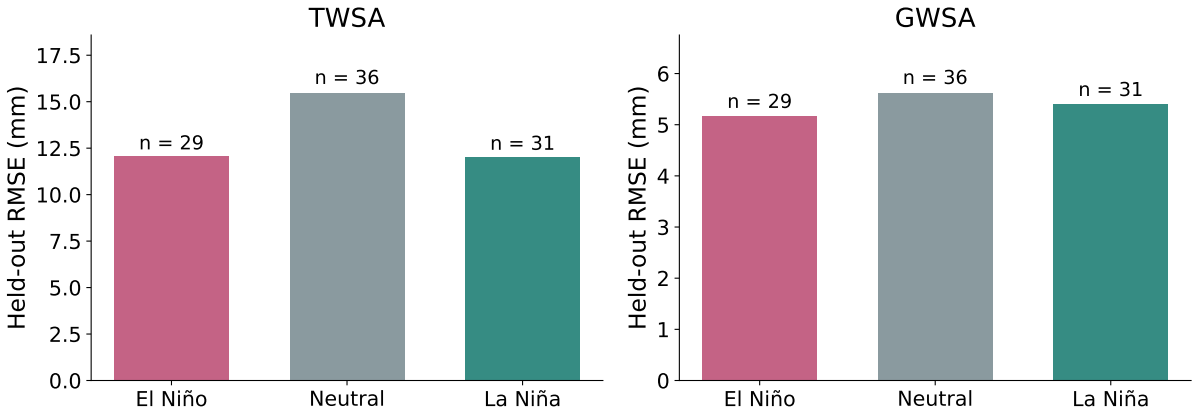
# 8 Results ENSO timing and storage




**ENSO timing in storage components**. National partial correlations remove season, trend and contemporaneous DMI, 2003–2023. All five selected ENSO leads are shown. The residual has a weaker six-month association, but this does not prove complete climate removal.

**Interpretation.** At a six-month ONI lead, the national climate-baseline association is about −0.53 for both storage variables, whereas residual correlations are about −0.14. This suggests a clearer descriptive ENSO signal in the baseline. No basin result passes the full conservative FDR screen; low power prevents concluding that climate leakage is absent.

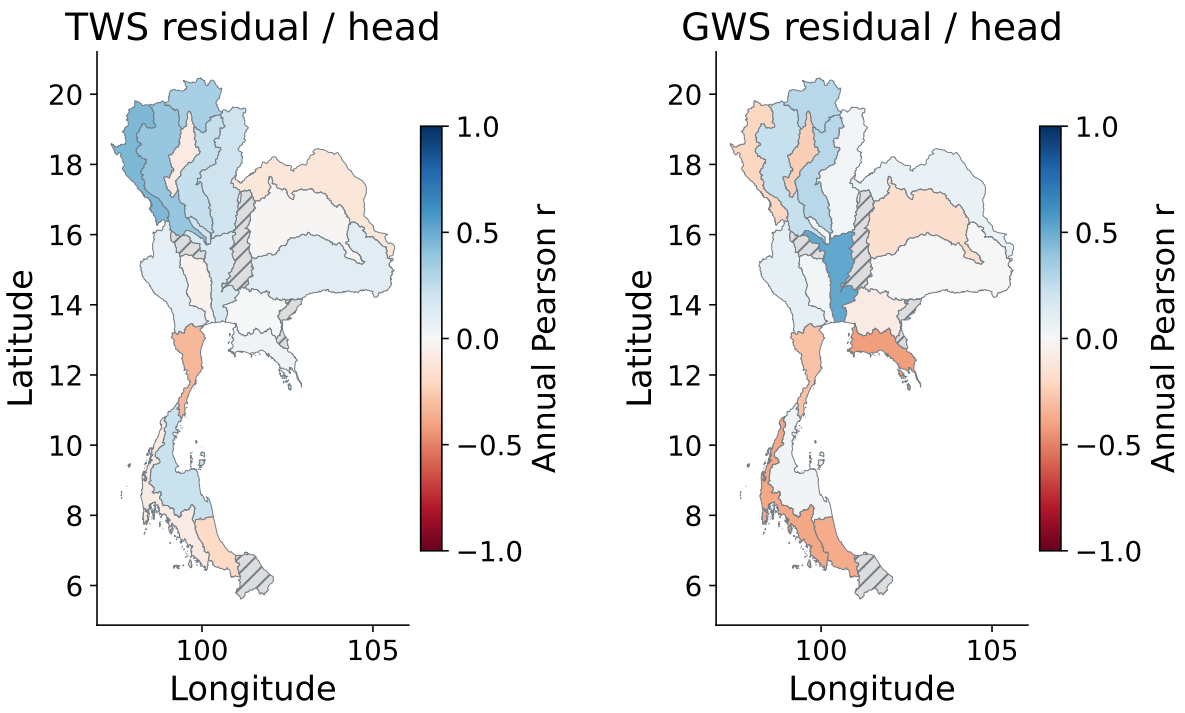
# 8 Results reconstruction during ENSO phases




**Model reconstruction during ENSO phases**. Held-out parent reconstruction, with 29 El Niño, 36 neutral and 31 La Niña months. Months share events; this is not independent event prediction.

**Interpretation.** Phase-conditioned reconstruction errors use a frozen model and 96 test months. Months share ENSO episodes and their target variances differ. The chart describes reconstruction under different climate conditions, not an independent forecast of El Niño impacts.

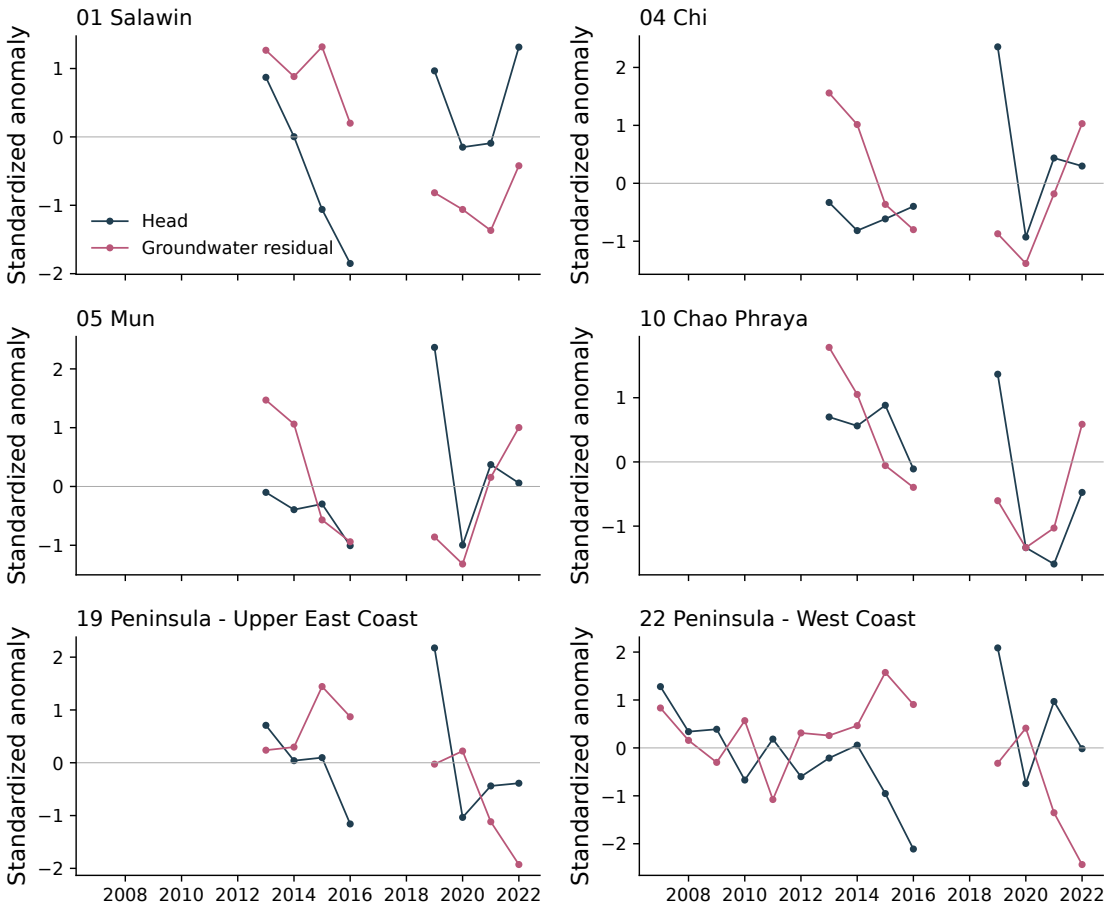
## 8 Results groundwater agreement in space




**Groundwater consistency and its spatial limits**. Annual residual–head correlations. Hatching indicates inadequate support. Head and basin storage have different units and observation scales; no storage-volume validation is implied.

**Interpretation.** National longest-overlap residual–head correlations are only 0.127 for total storage and 0.037 for groundwater. Basin evidence is uneven, and hatched areas lack adequate support. Local groundwater level is useful state evidence but cannot validate the whole human residual.

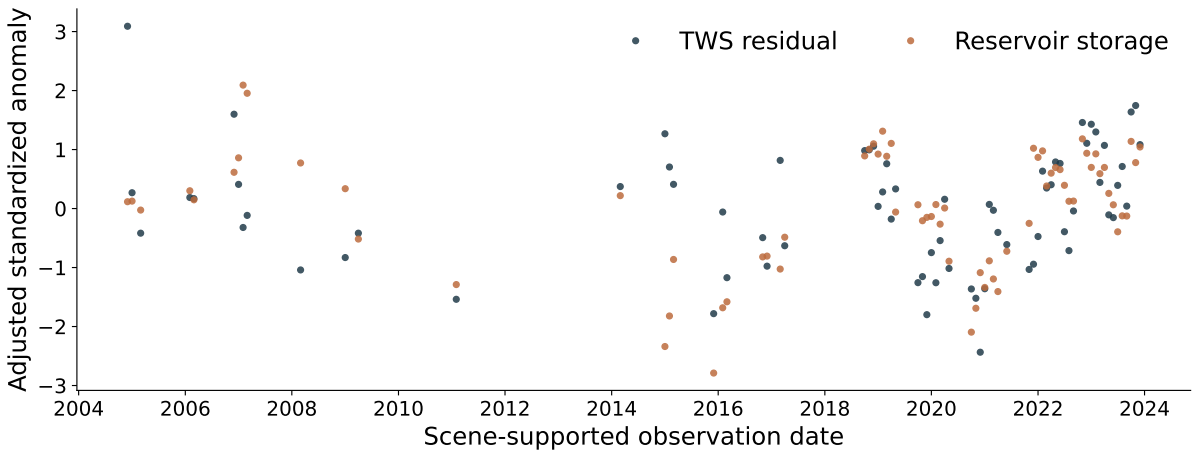
# 8 Results groundwater timing and network effects




**Groundwater levels and groundwater residual timing**. Six geographical examples on jointly available annual support. Standardization permits a timing comparison but does not convert head into groundwater volume. Short samples and changing stations limit inference; gaps remain missing.

**Interpretation.** Agreement changes with station selection and available years. In a strict balanced panel, national groundwater-residual correlation is approximately zero over eight usable years. The illustrated standardized curves compare timing; they do not convert metres of head into millimetres of storage.

# 8 Results independent reservoir evidence




**Independent reservoir component evidence**. Scene-supported values, with season and trend removed, then standardized. Points preserve observation gaps. Adjusted national r = 0.439 across 73 months; year-cluster 95% interval 0.188–0.665.

**Interpretation.** After season and trend removal, national reservoir storage correlates with the total residual at r = 0.439 over 73 scene-supported months, with year-cluster interval 0.188–0.665. This supports a surface-water component. Reservoir inflow and operating choices are mixed, so it is not human-only or groundwater validation.

## 8 Results independent evidence for every basin

| Basin | TWS r | TWS detr r | Years | GWS r | GWS detr r | Years |
|---|---|---|---|---|---|---|
| 01 | 0.449 | 0.582 | 9 | -0.215 | 0.167 | 8 |
| 02 | 0.330 | 0.345 | 17 | 0.279 | 0.399 | 16 |
| 03 | -0.121 | -0.084 | 9 | 0.063 | -0.109 | 8 |
| 04 | -0.009 | -0.030 | 9 | -0.168 | -0.023 | 8 |
| 05 | 0.113 | 0.103 | 9 | -0.007 | 0.070 | 8 |
| 06 | 0.385 | 0.432 | 15 | 0.229 | 0.293 | 14 |
| 07 | -0.089 | -0.043 | 9 | -0.244 | 0.047 | 8 |
| 08 | 0.236 | 0.157 | 9 | 0.277 | -0.124 | 8 |
| 09 | 0.208 | 0.176 | 9 | 0.026 | -0.198 | 8 |
| 10 | 0.152 | 0.050 | 9 | 0.519 | 0.200 | 8 |
| 11 | — | — | 5 | — | — | 4 |
| 12 | — | — | 5 | — | — | 4 |
| 13 | -0.044 | -0.212 | 17 | 0.027 | -0.147 | 16 |
| 14 | 0.094 | 0.074 | 17 | 0.093 | 0.116 | 16 |
| 15 | 0.001 | -0.041 | 9 | -0.082 | -0.187 | 8 |
| 16 | — | — | 3 | — | — | 2 |
| 17 | 0.041 | 0.136 | 9 | -0.420 | -0.215 | 8 |
| 18 | -0.335 | -0.394 | 9 | -0.289 | -0.115 | 8 |
| 19 | 0.227 | 0.096 | 9 | 0.033 | -0.147 | 8 |
| 20 | -0.197 | -0.656 | 18 | -0.375 | -0.722 | 17 |
| 21 | — | — | 0 | — | — | 0 |
| 22 | -0.080 | -0.143 | 15 | -0.389 | -0.450 | 14 |

Longest-overlap retained-flags panel. Positive r means lower residual tends to occur with lower head. Detr r removes a linear trend. An em dash means unavailable evidence, not zero. Head is not converted into storage; no cross-unit NSE or RMSE is reported. Other station panels and conditional intervals remain in the complete CSV.

**Interpretation.** The table exposes observation gaps as well as correlations. A basin with missing evidence is not assigned zero impact. Read head agreement together with paired years and reservoir support; neither measurement directly observes a human-only counterfactual.

# 9 Conclusion

The primary CSR-based analysis identifies a national decrease in total and derived groundwater storage, with seasonal Sen slopes of $-7.51$ and $-6.84$ mm/year on their respective records. Alternative processing retains a negative national direction but changes the estimated magnitude and basin ordering. The basin results are therefore most useful when viewed as broad screening evidence with explicit observational support.

Weather memory materially improves reconstruction of the naturalized WaterGAP target. Held-out ensemble scores are high, while anomaly metrics and individual years expose weaknesses. GLDAS adds a matched land-component benchmark. ENSO diagnostics describe lagged climate associations and phase-dependent reconstruction without providing a unique causal attribution. The residual remains $R = H + E$; independent wells are inconclusive and reservoirs support only a surface-water component.

The three objectives are delivered as basin trend products, an evaluated reconstruction-and-residual framework, and a reproducible decision-support website. The final release includes original publication figures, full-precision tables, frozen teleconnection inputs, executable code, a separate manuscript and an academic presentation. Monitoring priorities should follow coherent regional signals and gaps in evidence. Quantitative aquifer-management conclusions require additional pumping, aquifer-property and spatial-response information.

## 10 Discussion interpreting the evidence

The evidence is strongest for a national decline in the primary storage records and for the value of climate memory in reconstructing the chosen naturalized simulation. Evidence is weaker for fine basin rankings and for a quantitative human-only residual. This hierarchy is consequential: a model can closely reproduce another model while both differ from satellite observations, and a statistically clear residual trend can still contain substantial structural discrepancy.

The ENSO extension improves interpretation by asking whether large-scale climate variation remains visible after baseline subtraction. The relatively weak six-month residual associations are compatible with partial removal of this linear climate signal, but do not establish an exhaustive natural counterfactual. The conservative rotation screen and the number of available ENSO episodes constrain inference. A stronger future study would combine longer records, physically matched spatial responses, alternative naturalized models and independent management or abstraction histories.

The processing-chain comparison is especially relevant to basin ranking: common-date Spearman correlations are only 0.138 for total storage and 0.027 for groundwater. A stable national negative direction can therefore coexist with an unstable local ordering. The maps should guide where to investigate, not justify a precise local abstraction limit.

# 10 Discussion groundwater and human influence

Groundwater level provides a local state response and reservoirs provide a surface-water state response. They are complementary to GRACE but neither is a direct measurement of national human storage change. A conversion $\Delta GWS \approx Sy \times \Delta h$ requires representative specific yield, compatible aquifer thickness and area, and appropriate treatment of confined conditions. Those data are not available at the required basin scale here. Reporting a volumetric pumping validation from the present wells would therefore overstate the observations.

The proposed publication contribution is an attribution-credibility assessment across all principal basins, not the assertion that a new machine-learning method uniquely separates climate and human effects. Journal submission should follow advisor review of source licensing, final bibliography, spatial-response treatment and whether the scientific contribution fits the target journal. The reproducible negative and sensitivity results are part of that contribution; improving figure appearance alone cannot remove the underlying identification limits.

The practical distinction is between detecting a storage change and identifying its cause. Wells add local groundwater-state evidence; reservoirs add surface-water evidence. Together they strengthen interpretation of components, but do not close the national human water balance. Their limitations remain part of the result rather than a footnote to be ignored.

# 11 Further work improve observations

The first priority is a longer, consistent groundwater monitoring panel. Basin agencies should provide historical monthly levels with coordinates, aquifer and screen information, datum, observation dates and quality flags. Keeping the same station population through time is essential for distinguishing physical change from changes in the monitored network.

A defensible conversion from groundwater level to storage additionally needs representative specific yield or storage coefficient, aquifer extent, confinement and vertical connectivity. These properties should be measured or justified by aquifer, rather than assigning one generic value to every basin. Pumping, return flow and recharge information are needed to relate storage change to abstraction.

Reservoir evidence should be expanded using consistent scene-based volumes and documented operating records. River discharge, floodplain water and inundation extent can constrain additional surface-water terms. These observations should be kept independent of any products used to derive the quantity being evaluated.

These additions would improve the ability to distinguish groundwater depletion, surface-water management and climate-driven changes. The immediate output should be a monitoring-gap inventory and a documented common observation panel, rather than another unsupported percentage of human impact.

# 11 Further work strengthen attribution and use

A methodological priority is to match the effective spatial response of gravity and the subtracted compartments through a justified filtering or forward-model experiment. The effects of basin size and coastal leakage should be assessed before treating fine basin contrasts as independent local signals.

Alternative naturalized hydrological models would test whether the residual depends on the chosen counterfactual. Longer climate records and carefully selected ENSO/IOD alternatives could improve power, but test data must remain separate from model selection. Nonlinear and lagged climate responses should be evaluated with a fixed analysis plan and dependence-aware uncertainty.

The decision-support system should add new observations with versioned provenance, retain missing-data flags and display uncertainty with each basin result. An independent researcher should rerun the numerical pipeline and review the interpretations before the information is used for a management decision.

For publication, the next step is advisor review of the contribution, matched spatial support, source terms, final authorship and declarations. The useful contribution is a transparent basin assessment that distinguishes what is detected, what is modeled and what remains unresolved. Improving presentation helps communication; additional evidence is needed to strengthen causal attribution.

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